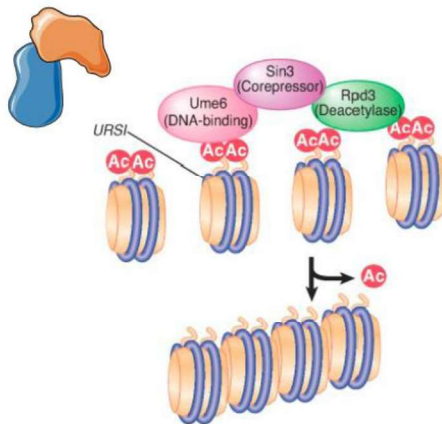


4. Chromatine remodeling

Histone Acetylation Is Associated with Transcription Activation

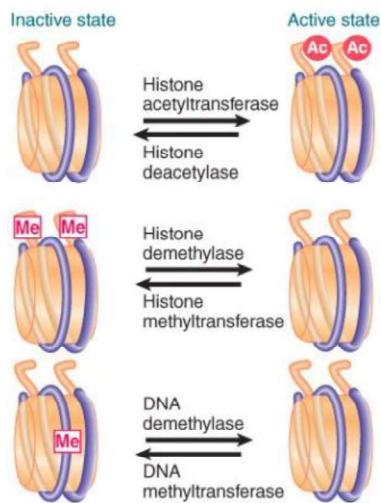


- Overall methylation vs acetylation



4. Chromatine remodeling

Methylation of Histones and DNA is connected



- **DNA methylation** is associated with **repression** of gene expression.
- **DNA de-methylation** is associated with **activation** of gene expression.
- **Histon methylation and de-methylation** is associated with both **activation** and **repression**. The specific response depends on every single case.
- DNA methylation is kept upon replication. They are involved in epigenetics.

- Methylation and acetylation is the most important chromatin remodeling (but there is many others like phophoylation)

Which of the following statements about chromatin remodelling is TRUE

Histone acetylation is associated with transcription repression

0%

Histone de-acetylation is associated with transcription activation

0%

Histone methylation is not associated with transcription activation

0%

All remodelling complexes contain an ATPase catalytic subunit

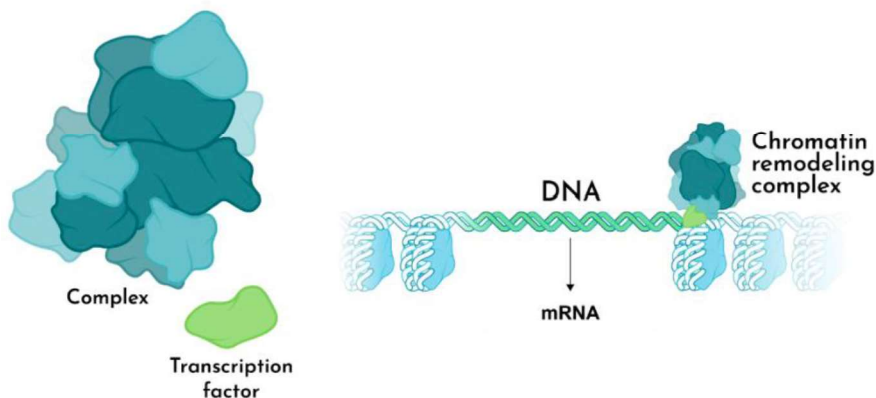
0%

- Answer: All remodelling complexes contain an ATPase catalytic subunit

Activators can only recognise their target on open chromatin

- ANSWER: FALSE
- Some activators are made to recruit the chromatin remodelling complexes needed to open the chromatin.

Which element is carrying the DNA-specificity in the following image representing the recruitment of a histone remodelling complex?



- Chromatin remodeling complexes are blind – they will do their job if they are recruited so either by the transcription factors or the transcription factor binding the chromatin remodeling complex by the histones

Can you identify which is the specific chromatin remodelling complex activity in the image

Histone acetyltransferase (HAT)

0%

Histone methyltransferase (HMT)

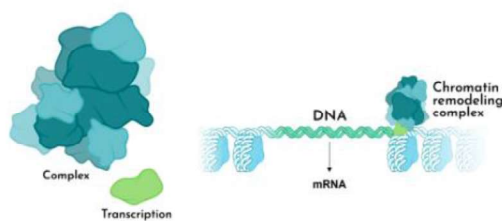
0%

Histone acetylase or histone methyltransferase

0%

Histone acetyltransferase (HAT) or Histone methyltransferase (HMT)

0%



- ANSWER: Histone acetyltransferase (HAT) or Histone methyltransferase (HM)

5. Activators and repressors



5. Activators and repressors

True activators: positive control

• Activator/repressor

- **An Activator** determine the frequency of transcription.
 - Activators work making protein-protein contacts with the basal factors
 - Activators may work via coactivators
 - Activators are regulated in many different ways
- **A Repressor** – A protein that inhibits expression of a gene.
 - Repressors may act to prevent transcription by binding to an operator site in DNA
 - Repressors can recruit chromatin remodelling complexes
 - Repressors can prevent translation by binding to RNA. **ONLY used in eukaryotic systems**

Types of activators

- **True activators**
 - Classical transcriptions factors that make **physical contact with the basal apparatus** directly or through and activator.
- **Antirepressors**
 - Recruit **chromatin-remodelling complexes**
- **Architectural protein**
 - These proteins function **bendind the DNA**, either bringing bound proteins closer to facilitate a cooperative complex or the other way around.

Types of genes

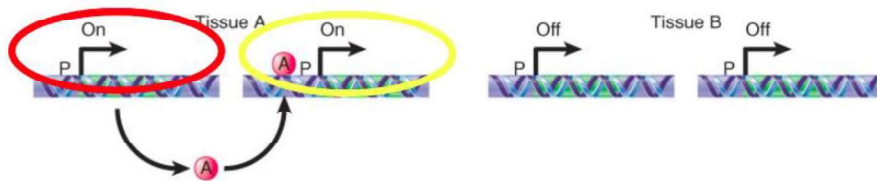
- **Positive control:** The default state of genes that are under positive control is that they cannot be expressed unless a positive regulator is bound.
- **Negative control:** the default state of genes that are under negative control is to be expressed. A specific intervention is required to turn them off.

- True activators, antirepressors, architectural protein, positive control, negative control, activator, repressor.
- **Positive control**



5. Activators and repressors

True activators: positive control



- **Positive control:** The default state of genes that are under positive control is that they cannot be expressed unless a positive regulator is bound.

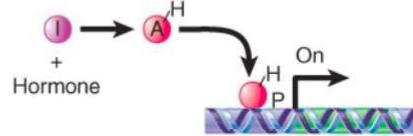
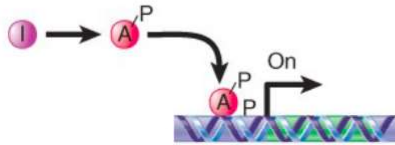
- **Why to regulate gene expression**
 - Tissue differentiation
 - Environmental response

- **Regulator gene:** A gene that encodes a product (typically protein) that controls the expression of other genes (usually at the level of transcription).
- **Structural gene:** A gene that encodes any RNA or protein product other than a regulator.

- Regulator gene and structural gene
- Positive control: The default state of genes that are under positive control is that they cannot be expressed unless a positive regulator is bound.

5. Activators and repressors

True activators: positive control

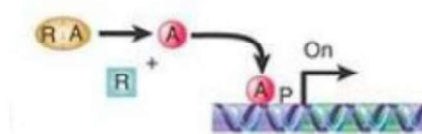
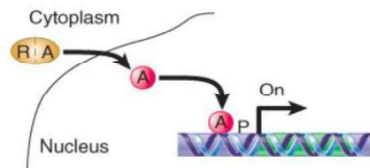


- **Covalent modification** e.g., phosphorylation
- It may be a kinase that become activated by an **environmental signal**
- **Binding of a ligand** e.g., hormone – functions as a co-activator
- i.e., an allosteric effector modifies the structure of the activator

- Why to regulate gene expression
 - Tissue differentiation
 - Environmental response

5. Activators and repressors

True activators: positive control



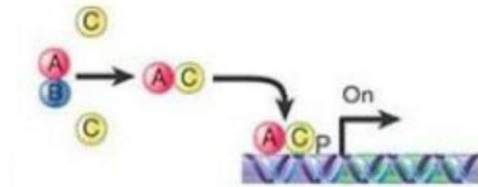
- **Sequestering in the cytoplasm** by protein-protein binding
- or activation by proteolysis

- Why to regulate gene expression
 - Tissue differentiation
 - Environmental response

- True activators: Repressors and activators are in the cytoplasm and the gene that they need to activate or repress is in the nucleus – so they need to be able to cross from the cytoplasm into the nucleus

5. Activators and repressors

True activators: positive control



- Why to regulate gene expression
 - Tissue differentiation
 - Environmental response

- Correct binding partner for activation
- Common on helix-loop-helix proteins

- Correct binding partner for activation
- Negative control

5. Activators and repressors

True activators: positive control

• Activator/repressor

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 - Activators work making protein-protein contacts with the basal factors
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Types of activators

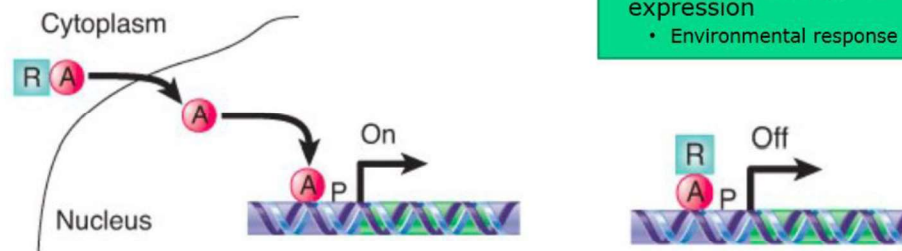
- **True activators**
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 - Recruit **chromatin-remodelling complexes**
- **Architectural protein**
 - These proteins function **bending the DNA**, either bringing bound proteins closer to facilitate a cooperative complex or the other way around.

Types of genes

- **Positive control:** The default state of genes that are under positive control is that they cannot be expressed unless a positive regulator is bound.
- **Negative control:** the default state of genes that are under negative control is to be expressed. A specific intervention is required to turn them off.

5. Activators and repressors

True activators: negative control



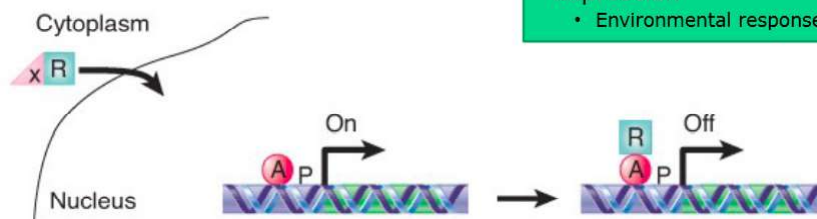
- Why to regulate gene expression
 - Environmental response

- **Sequestering the activator** by protein-protein binding **with a repressor**

- True activators negative control

5. Activators and repressors

True activators: negative control



- Why to regulate gene expression
 - Environmental response

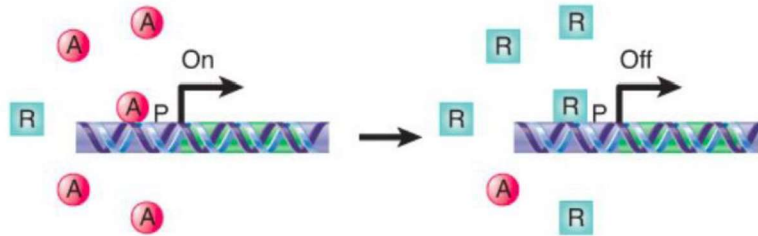
- **Sequestering the activator** by protein-protein binding **with a repressor**
- It can **repress the expression only in the absence of an inducer**
- Alternatively, **repressor binding can be dependent on a co-repressor**

- Competition between an activator and repressor for the same binding site

5. Activators and repressors

True activators: negative control

- Why to regulate gene expression
 - Environmental response



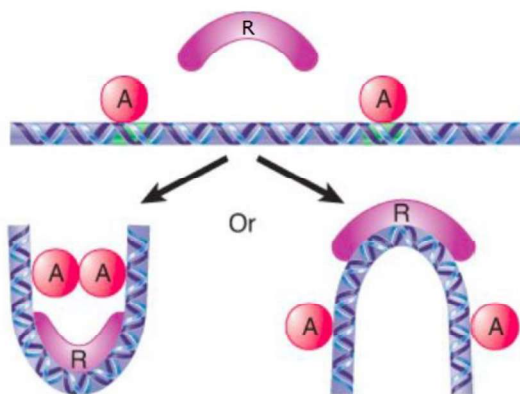
- Competition between an activator and repressor for the same binding site.

- Architectural proteins



5. Activators and repressors

Architectural proteins



- **Architectural protein** – A protein that, when bound to DNA, can alter its structure (e.g., introduce a bend).
- Architectural proteins **control the structure of DNA** and thus control whether **bound proteins can contact each other**. They can either facilitate or prevent protein-protein contact.

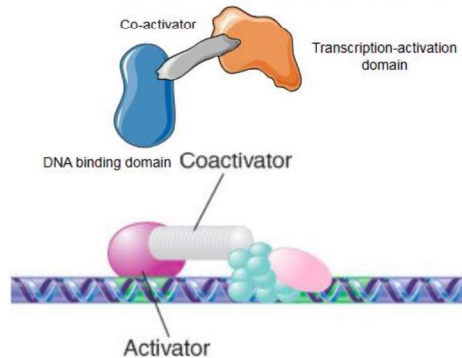
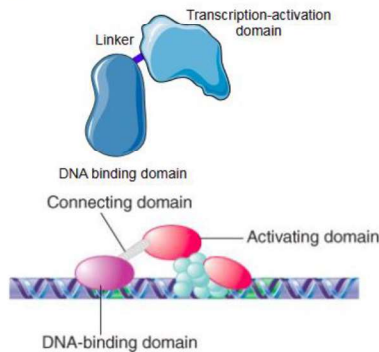
- The most important thing is that the RNA-polymerase is assembled around the promoter

5. Activators and repressors

Activators Interact with the Basal Apparatus



- DNA binding and transcription-activation activities are carried out by independent domains of an activator
- The role of the DNA-binding domain is bringing the transcription-activation domain into the vicinity of the promoter
- An activator that does not have an activating domain may work by binding a coactivator that has an activating domain.
- Several factors in the basal apparatus are targets with which activators or coactivators interact.



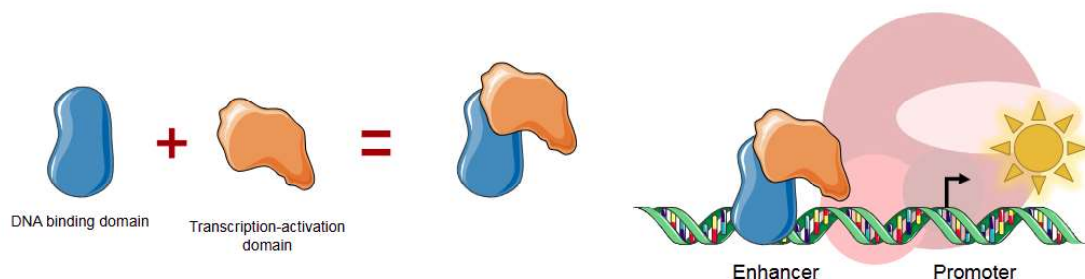
- DNA binding domain co-activator – sometimes transcription activations domains don't make direct contact to the DBD but binds to an intermediate protein called a co-activator.
- You always need both, but there can be intermediate proteins – but you need to have the chain that connects the DNA with the ____

The Two-Hybrid Assay Detects Protein–Protein Interactions

5. Activators and repressors

The Two-Hybrid Assay Detects Protein–Protein Interactions

- The two-hybrid assay works by requiring an interaction between two proteins, where one has a DNA-binding domain and the other has a transcription-activation domain

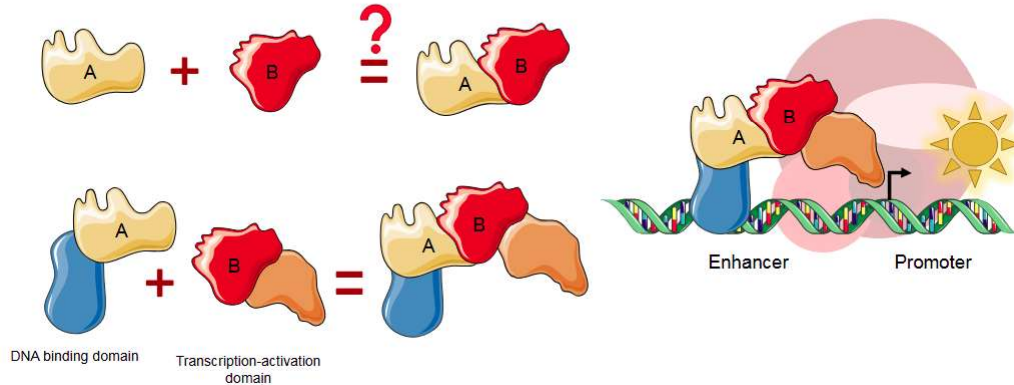


- Check if two proteins are interacting – a lab experiment to check this – the DBD and the transcription activation domain needs to somehow come together for the expression to occur. You can measure a function of how well the two proteins interact.

5. Activators and repressors

The Two-Hybrid Assay Detects Protein–Protein Interactions

- The two-hybrid assay works by requiring an interaction between two proteins, where one has a DNA-binding domain and the other has a transcription-activation domain



- If A and B can come together it will be equivalent to a __ activator as described before – if they come together then they will make a signal.

- Case with The Two-Hybrid Assay

Discover which proteins interact with BCRA1

- Identify proteins interacting with BRCA1 using yeast two-hybrid (Y2H) assay and determine the effects of these interactions on apoptosis and cell proliferation in mammalian cell cultures.



Protein Candidate	α -Galactosidase Activity (Weak/Strong)	Apoptosis (Increase/Decrease/No Change)	Proliferation (Increase/Decrease/No Change)
Protein A	Strong	Increase	Decrease
Protein B	N/A	N/A	N/A
Protein C	Weak	No Change	No Change
Protein D	Strong	Decrease	Increase
Protein E	N/A	N/A	N/A

α -Galactosidase Activity (Weak/Strong): Indicates the level of α -galactosidase activity, reflecting interaction strength.

Apoptosis (Increase/Decrease/No Change): Indicates the observed change in apoptosis levels in mammalian cells overexpressing the candidate protein.

Proliferation (Increase/Decrease/No Change): Indicates the observed change in cell proliferation in mammalian cells overexpressing the candidate protein.

- First identify which proteins interact together – and then identify any of these proteins that can be identified in cancer (good proteins to try to attack in cancer)
- Protein A, C and D is interacting with the BRCA1
- The effect of these interactions is for C, no change, but for A: ?
- D could lead to cancer – good candidate to look at if you want to detect cancer

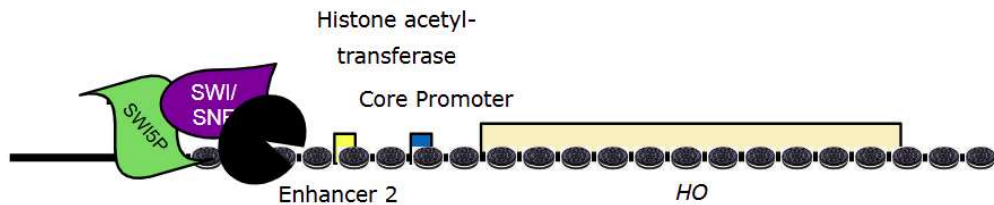
6. Examples:

- HO genes

6. Examples

HO gene activation in yeast

- The *HO* gene in yeast is involved in mating type shift
- The activation requires chromatin remodeling prior to activation
- **SWI5P and SBP are activators binding at two different enhancer sites (aka Upstream Activating Sequences = UAS)**
- **SWI5P recruits proteins required for histone removal**
- SBP recruits TFIID – and subsequent RNA Pol II to the promoter
- The *HO* gene is transcribed



- HO gene activation in yeast
- SBP – true activator
- SWI5P – activator

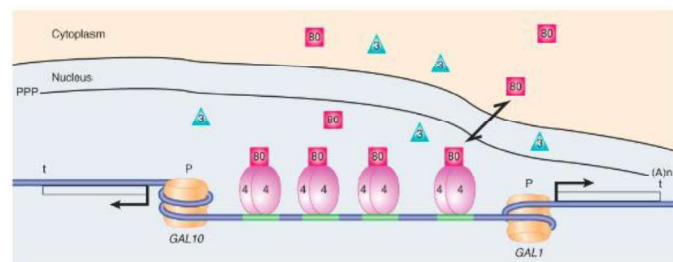
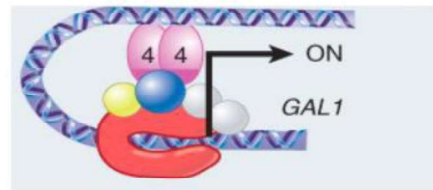
- GAL genes



6. Examples

Yeast GAL Genes: required for utilization of galactose

- **Gal4 Recruits TF_{II}D to the promoter and activates transcription – ACTIVATOR**
- **Gal80 sequesters the interaction between Gal4 and TF_{II}D – REPRESSOR**
- Gal3 binds to Gal80 and prevents the interaction with Gal4 – REPRESSOR OF THE REPRESSOR → ACTIVATOR
- Gal3 only binds to Gal80 in the presence of galactose – ACTIVATOR DEPENDENT ON LIGAND
- Additional regulatory circuits are involved



- Yeast GAL genes: required for utilization of galactose
- Gal4, Gal80 and Gal3

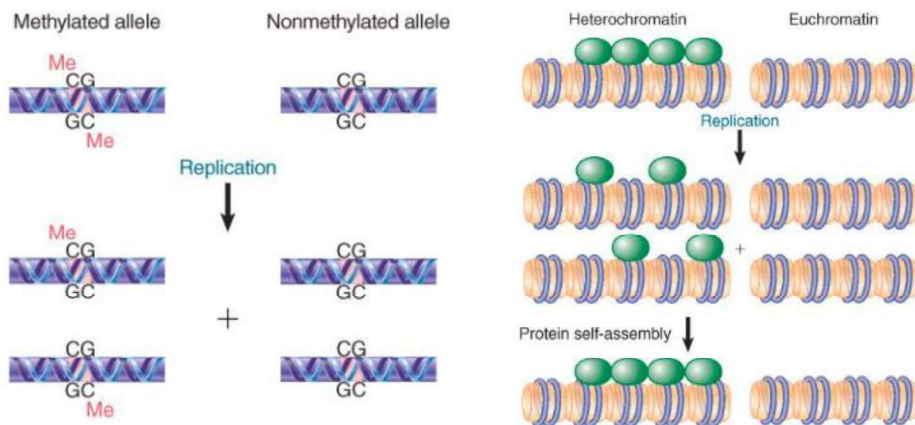
7. Epigenetics

Epigenetic effects can result from modification of a nucleic acid after it has been synthesized or by the perpetuation of protein structure. Hereditary features that are not encoded on the DNA. Things that is going to pass from generation to generation.

Most common epigenetics is methylation of the DNA:

7. Epigenetics

Modification of nucleic acids after synthesis



- The methylation pattern is inherited as hemi-methylated DNA is subject to methylation, but unmethylated is not
- The chromatin structure is inherited.
- Makes so something can or cannot be expressed – they have an impact on gene expression. Methylated allele is not expressed and in replication the molecule itself will now be modified to be methylated.

7. Epigenetics

Examples

Queen bees



A new **queen honeybee** is born from an egg laid by an existing queen. The egg hatches into a larva, which is then fed a special diet by nurse bees in order to become sexually mature. Thirteen days after the egg hatches into a larva, a new queen bee emerges. Now **researchers have found significant differences in the methylation patterns of more than 550 genes** — most of which are involved in essential cellular activities such as metabolism and RNA synthesis.

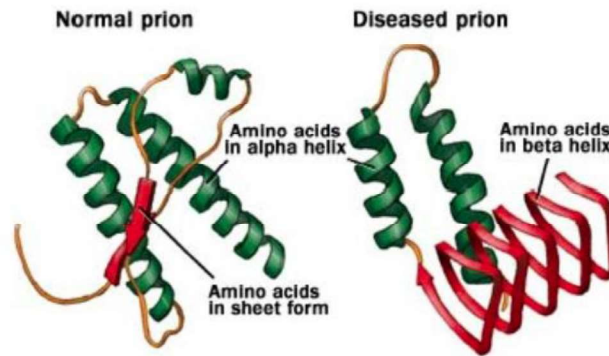
Calico cats



The **calico cat** is an example of **X-chromosome inactivation**. Calico cats are always females. They appear as a mixture of red and black colored fur, depending on which X chromosome has been inactivated. **The inactivated chromosome is heavily methylated.**

- Perpetuation of protein structures, prion

- **Prion** – A proteinaceous infectious agent that behaves as an inheritable trait, although it contains no nucleic acid.
 - The diseased prion induces a shift in tertiary structure from a normal structure into the diseased structure
 - Examples are PrP^{Sc}, the agent of scrapie in sheep and bovine spongiform encephalopathy.



- By getting unfolded It causes the other proteins to get unfold in the same manner – leading to difference in expression

Which statement regarding Activators is TRUE

True activators can establish protein-protein interaction with TFIID

0%

Activators require covalent modification

0%

Activators avoid the action of repressors

0%

Activators always require co-activators to do their function

0%

- ANSWER: True activators can establish protein-protein interaction with TFIID
- The others are also true, but not ALWAYS

Which statement regarding eukaryotic Activators is NOT true

Activators prevent transcription by binding to an operator site

0%

True activators establish protein-protein contact with the basal apparatus

0%

Activators may work via co-activators

0%

Activators can recruit histone acetylases

0%

- ANSWER: Activators prevent transcription by binding to an operator site

These are the results of a two-hybrid assay. The objective is to identify is A1 and A2 can interact with B1 and B2. Which controls are missing to interpretate the results?

Protein A (conjugated with DNA-binding domain)	Protein B (conjugated with Activation domain)	Relative β -galactosidase Activity
A1	B1	100%
A1	B2	15%
A2	B1	5%
A2	B2	80%

- ANSWER: The control missing to interpretate the results is the proteins alone – A1, A2, B1 and B2 alone to make sur that they cannot interact alone and that there is no observed activity for them alone.

Molecular Biology Week 7: Regulatory RNA's

Learning objectives

After today's lecture and exercises, a student who has met the objectives will be able to:

- Predict secondary structures in simple RNA molecules.
- Describe the principle of mutually exclusive structures
- Describe the role of the following common RNA classes in cell function: mRNA, tRNA, ncRNA, miRNA, siRNA, asRNA
- ncRNA, antisense RNA, miRNA, and siRNA and their abundance.
- Describe the attenuation mechanism
- Describe how metabolites can stabilize a specific RNA structure (ie riboswitches).
- Describe the principal mechanisms by which RNA structure can regulate gene expression and identify terminator and antiterminator structures
- Explain the mechanism of a specific RNA-mediated regulatory system based on a figure.
- Analyze the effect of a specific mutation in a specific RNA-mediated regulatory system.

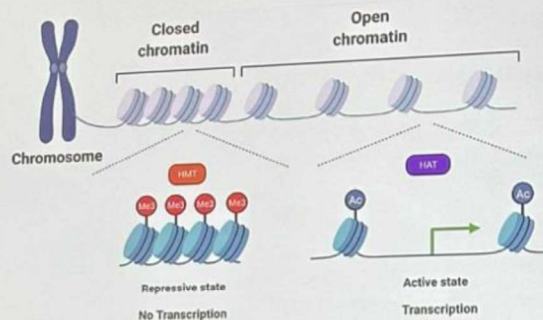
Notes from the lecture

Repetition for last time

Chromatin remodelling: acetylation, methylation and phosphorylation

Chromatin remodelling

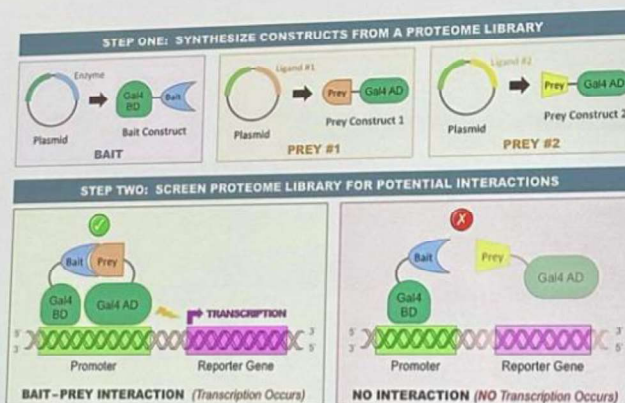
- In eukaryotes the DNA is assembled around proteins called histones that form octamers (nucleosomes)
- To express a gene the basal apparatus (TFIID+RNApolII) needs access to the DNA
- Histone remodelling complexes modify amino acids in the histones by adding/removing acetyl, methyl, or phosphorous groups modifying the chromatin structure and making it more tightly bound or loose to the DNA
 - Acetylation → activation
 - Methylation → activation or repression
 - Phosphorylation → regulatory functions



Yeats two hybrid assay : to check if two proteins are interacting or not

Yeast Two Hybrid assay

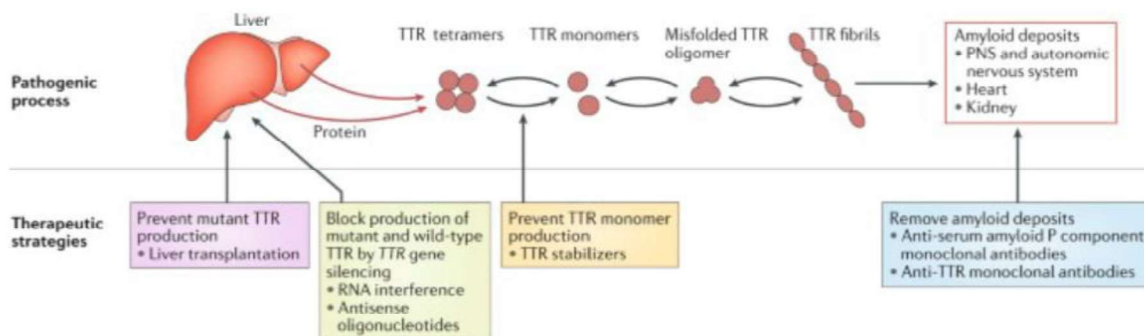
- It is an assay to determine where two proteins are interacting based on the modular nature of transcription factors
 - The bait is fused to a DNA binding domain that can bind to the promoter.
 - The prey is bound to a protein domain that interacts with the basal apparatus
 - The promoter is fused to a reporter gene
 - If prey and bait interact, the activity of the reported gene is measured



1. Introduction

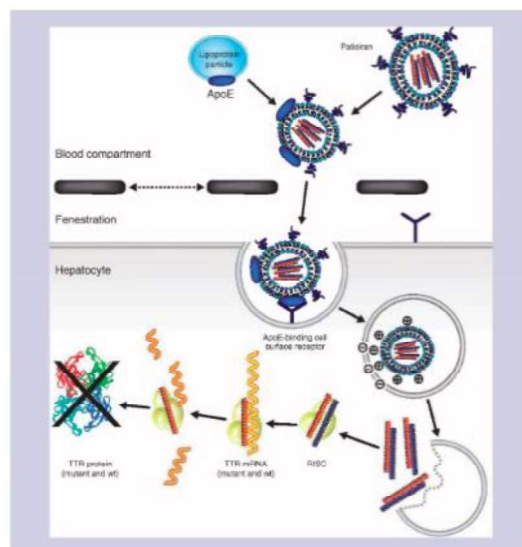
How can diseases in which mutations cause protein misfolding be treated?

- Hereditary Transthyretin Amyloidosis" (hATTR). hATTR is caused by mutations in the TTR gene, producing a misfolded protein, transthyretin (TTR), which accumulates in tissues and causes various symptoms.



“how it works”: RNA interference can reduce the accumulation of amyloids by reducing translation

- An interference RNA complementary to the coding sequence of the mutated TTR protein is encapsulated in lipid nanoparticles that are directed to the hepatocytes (apoE)
- The RNA interference is released from the lysosomal compartment at low pH
- The RNA interference binds to the mRNA of TTR, avoiding its translation
- The concentration of TTR is reduced and therefore the amyloid accumulation



Which of the following RNAs are not involved in the regulation of transcription or translation?

asRNA

miRNA

RNAi

Leader

Riboswitch

mRNA

rRNA

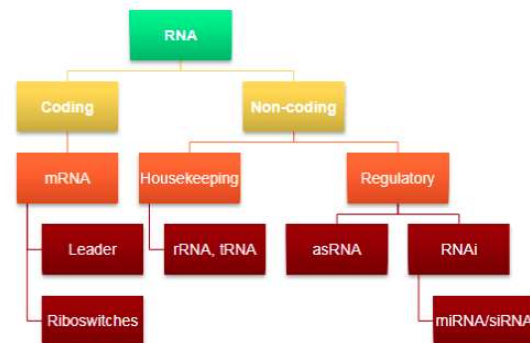
tRNA

siRNA

- ANSWER: They are all – in one way or another – involved in the regulation in one of the two processes somehow. rRNA is the most “right” one here if you really needed to choose one
- **Regulation by RNA**

RNA classification:

- RNA can function as a regulator by forming a region of secondary structure (either inter- or intramolecular) that can control gene expression.
- Vast tracts of the eukaryotic genome are transcribed on both strands.
- The list of different RNAs is huge – **do not try to learn them all by heart.**
- These are the important names to remember:
 - mRNA, tRNA, rRNA, **asRNA (antisense RNA)**.
 - **RNAi** aka silencing RNA or interfering RNA
 - **siRNA** (small interfering RNA) highly specific for 1 target
 - **miRNA** (micro RNA) multiple targets



- Housekeeping – ribosomal RNA and transfer RNA
- Regulatory - antisense RNA, RNA interference, micro RNA and ?RNA
- **Abundance would be the housekeeping ones followed by the mRNA, then the regulatory ones.**

Generally about antisense RNA:

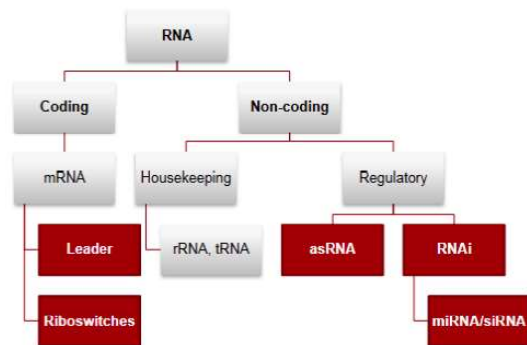
General function of antisense RNA

Antisense RNAs are RNA molecules that are complementary to messenger RNAs (mRNAs). Because of this complementarity, they can bind to the mRNA and change how it behaves. One common effect is that they **block the ribosome from attaching** to the mRNA; when this happens, the protein cannot be produced, so the gene is effectively silenced. Antisense RNA can also **interfere with RNA processing steps** such as splicing or polyadenylation, which means the mRNA is not shaped correctly and cannot be used to make a proper protein. In other cases, the binding of antisense RNA makes the mRNA **unstable and more likely to be degraded**, so the message is destroyed before any protein can be produced. Finally, antisense RNA can sometimes **hybridize with the DNA template itself**, which prevents transcription from continuing, meaning the mRNA is never even made in the first place.

Why they are needed

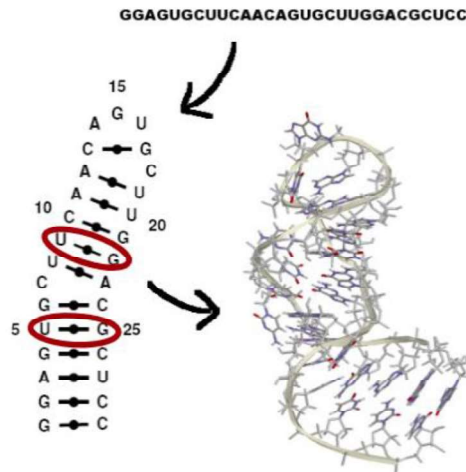
This form of regulation is important because it allows the cell to quickly and reversibly adjust gene expression without changing the DNA. By turning down or turning off specific mRNAs, the cell saves resources, avoids producing harmful amounts of proteins, and responds more flexibly to changing conditions. In short, antisense RNAs are needed because they provide the cell with an **efficient, fine-tuned way of controlling which proteins are made and when.**

- RNA can function as a regulator by forming a region of secondary structure (either inter- or intramolecular) that can control gene expression.
- Vast tracts of the eukaryotic genome are transcribed on both strands.
- The list of different RNAs is huge – **do not try to learn them all by heart.**
- These are the important names to remember:
 - **mRNA**, tRNA, rRNA, **asRNA (antisense RNA)**.
 - **RNAi** aka silencing RNA or interfering RNA
 - **siRNA** (small interfering RNA) highly specific for 1 target
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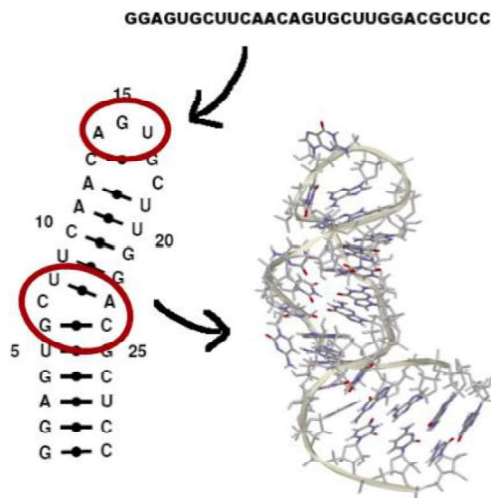
2. RNA structure

- RNA is a single strand molecule, that can form internal base pairing.
- RNA adopts secondary and tertiary structures by internal base pairing.
- **Some rules about stability:**
- U-G basepairs do not contribute to the stability of a stemloop structure, but is not preventing its formation



- The bases are not free – why it always has this shape – in RNA the bases are available to make hydrogenbonds with other molecules or itself.

- RNA is a single strand molecule, that can form internal base pairing.
- RNA adopts secondary and tertiary structures by internal base pairing.
- **Some rules about stability:**
- U-G basepairs do not contribute to the stability of : stemloop structure, but is not preventing its formation
- At least three nucleotides in the loop!
- Bulges in double-stranded RNA are allowed



The structure of the RNA is not going to be straight – it can have a very funky form on the secondary structure – tertiary structure of RNA would be when the RNA is interacting with other molecules.

- RNA folding server – how do we know the shape of the RNA

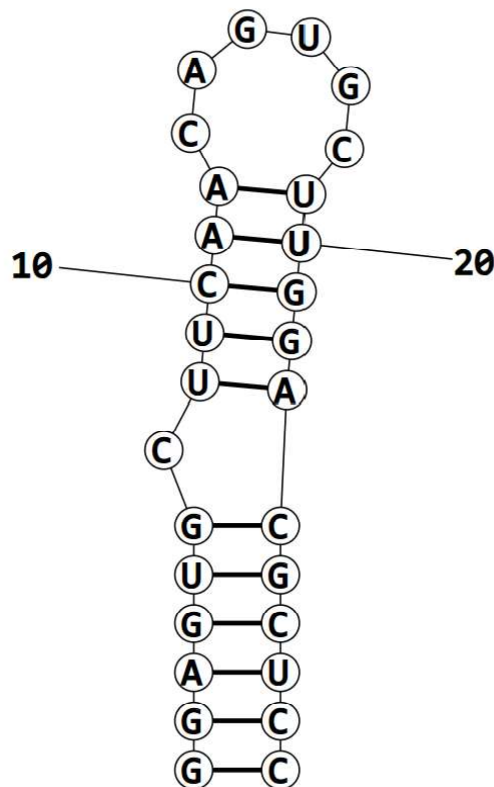
<https://rna.urmc.rochester.edu/RNAstructureWeb/Servers/Fold/Fold.html>

and put in the sequence: 5'-GGAGUGCUUCAACAGUGCUUGGACGCUCC-3

the server gives us the most stable possible secondary structure:

RNAstructure Fold Results

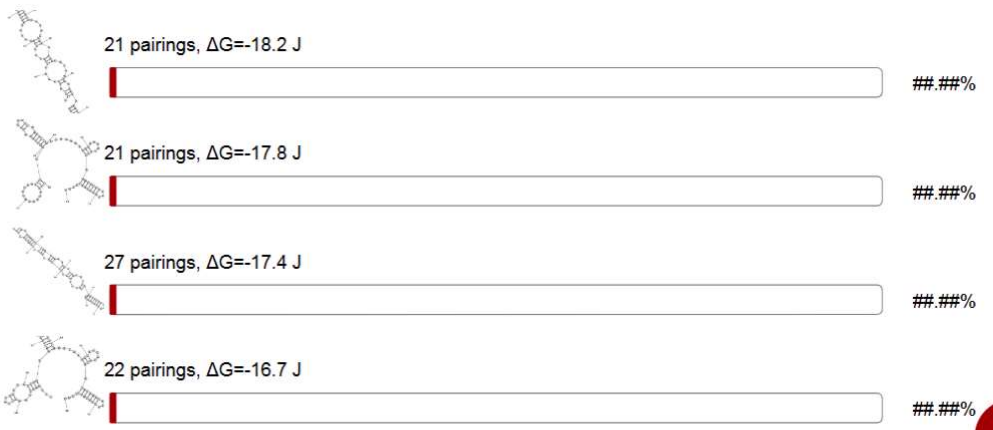
□ (Predict the lowest free energy structure and a set of low free energy structures for a sequence.)



ENERGY = -13.0 10/20/25 07:28:17

Download Structure: SVG JPEG PNG PDF PS CT

an online server and receive different folding possibilities with different scores. Which one is the most stable?



- ANSWER: the first one - The lowest delta G
- Shifting from one formation to another can happen with outside stimuli.

- RNA structure – a real life case



2. RNA structure RNA folding server: a real life case



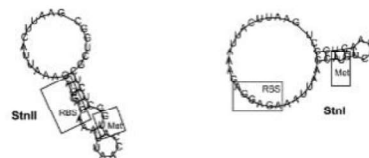
Journal of Biotechnology
Volume 127, Issue 2, 1 January 2007, Pages 211-221



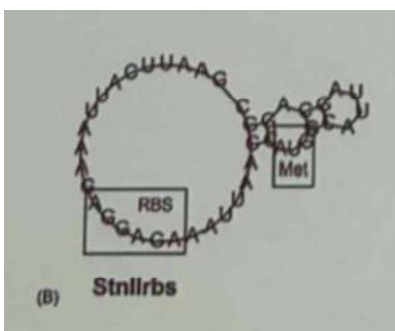
Silent mutations at the 5'-end of the cDNA of actinoporins from the sea anemone *Stichodactyla helianthus* allow their heterologous overproduction in *Escherichia coli*

Jorge Alegre-Cebollada, Georgia Clementi, Michela Cunliffe, Christian Porres, Mercedes Ofaderra, José G. Gavilanes, Álvaro Martínez del Pozo

Sequence alignment of actinoporin cDNA variants. The RBS (Ribosome Binding Site) is highlighted in the 5'-end of the cDNA. The StnII and StnI sequences are shown. The RBS is located at the 5'-end of the cDNA, and the StnII and StnI sequences are located at the 3'-end of the cDNA.



- They were trying to express sea anemone toxins in bacteria
- The expression of StnII was exceptionally low yield
- The mRNA took a secondary structure that hid the Ribosome Binding site
- By introducing silent mutations in the coding region, the yield of the purification increased 200x



RBS : ribosome binding site – one had a completely exposed and the other is two stranded – they made the one above and observed very good results afterwards – both Met and RBS are exposed. – so knowing the structure is very important when working with it.



2. RNA structure



GGAGUGCUUCAACAGUGCUUGGACGCUCC

A pilot study on the prevalence of DNA palindromes in breast cancer genomes

Sandeep Subramanian, Sriakshmi Chaparala, Viji Avalli & Madhavi K. Ganapathiraju

BMC Medical Genomics 9, Article number: 73 (2016) | [Cite this article](#)

2536 Accesses | 5 Citations | [Metrics](#)

Abstract

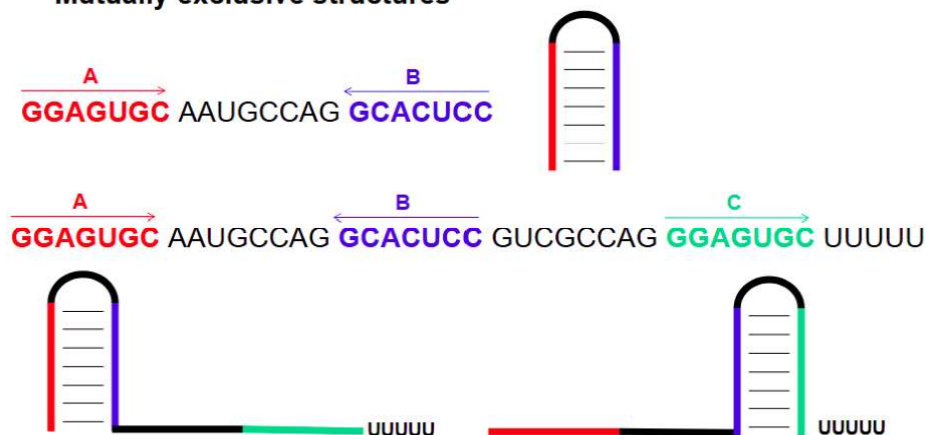
Background

DNA palindromes are a unique pattern of repeat sequences that are present in the human genome. It consists of a sequence of nucleotides in which the second half is the complement of the first half but appearing in reverse order. These palindromic sequences may have a significant role in DNA replication, transcription and gene regulation processes. They occur frequently in human cancers by clustering at specific locations of the genome that undergo gene amplification and tumorigenesis. Moreover, some studies showed that palindromes are clustered in amplified regions of breast cancer genomes especially in chromosomes (chr) 8 and 11. With the large number of personal genomes and cancer genomes becoming available, it is now possible to study their association to diseases using computational methods. Here, we conducted a pilot study on chromosomes 8 and 11 of cancer genomes to identify computationally the differentially occurring palindromes.

- Inverted repeats in RNA suggests the presence of a stem-loop structure
- What do inverted repeats in DNA suggests what (apart from the putative RNA stem-loop structure)?

... a binding site for a protein

Mutually exclusive structures



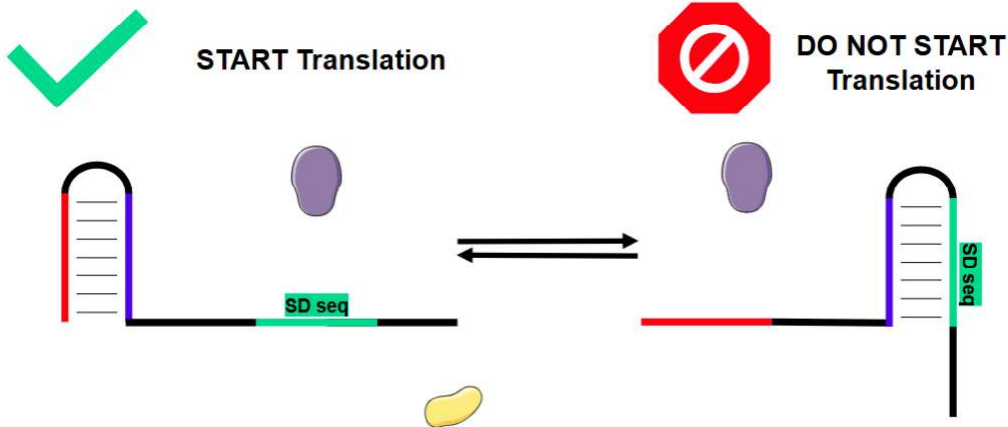
- B is both complementary with A and C so there are two possible shapes:
- 1) One of the structures is more thermodynamically stable

- 2) External stimuli can modify the equilibrium and provoke the swift between such structures.

The formation of one structure prevents the formation of another



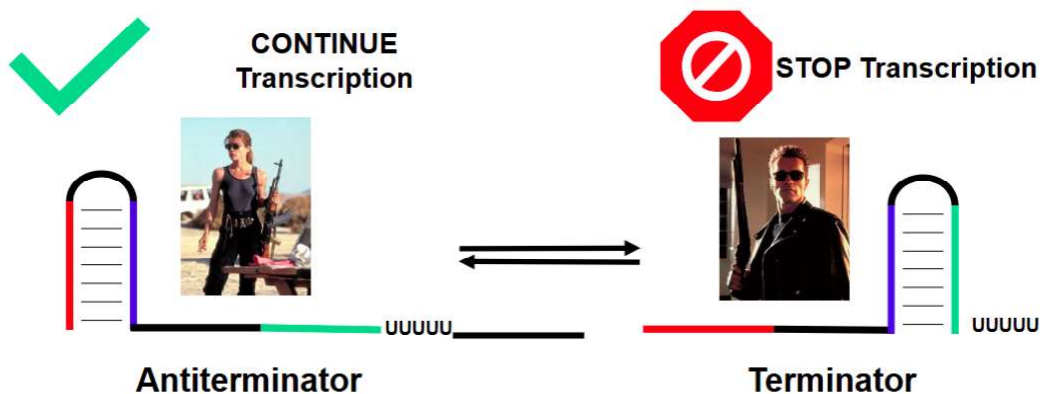
Mutually exclusive structures: sequestering the SD sequence



DTU

2. RNA structure

Mutually exclusive structures: terminator vs antiterminator



Antiterminator

Terminator

Terminator is a signal read by the RNA polymerase!

- Something about the rna-polymerase that if it binds to one or another it affects the transcription in one of the directions.
- RNA structure: Termination of the translation in prokaryotes

- Factor independent termination aka “Intrinsic terminators”
- A stem-loop in the RNA followed immediately by a stretch of U residues
- In the DNA sequence they can be identified by an inverted repeat immediately by a stretch of T residues

AGTAACGGTAGGCTCCCCTCTGGGGAGCCTTTTTTTTTTCCC
ACATTGCCATCCGAGGGGAGACCCCTCGGAAAAAAAAAAGGG



- The leader is the piece of the RNA that goes before the coding region

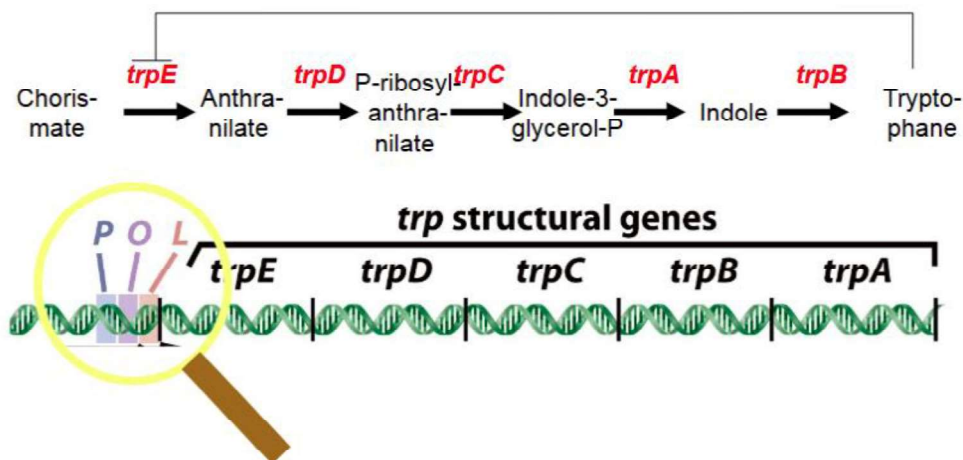
3. Gene regulation by RNA structure

- Attenuation

Example: The Tryptophane attenuator in *Escherichia coli*

Attenuation – Transcription Termination/Antitermination

Example: The Tryptophane attenuator in *Escherichia coli*



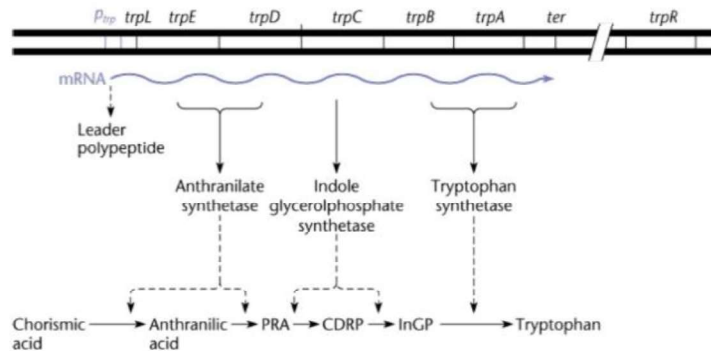
- Er are in bacteria (prokaryotes) so this is and operon, promotor, operator and leader (POL)

3. Gene regulation by RNA structure

Attenuation – Transcription Termination/Antitermination

Example: The Tryptophane attenuator in *Escherichia coli*

- The *trp* operon is negatively controlled by the level of its product, the amino acid tryptophan (**autoregulation**).
- The amino acid tryptophan activates an inactive repressor encoded by *trpR*.



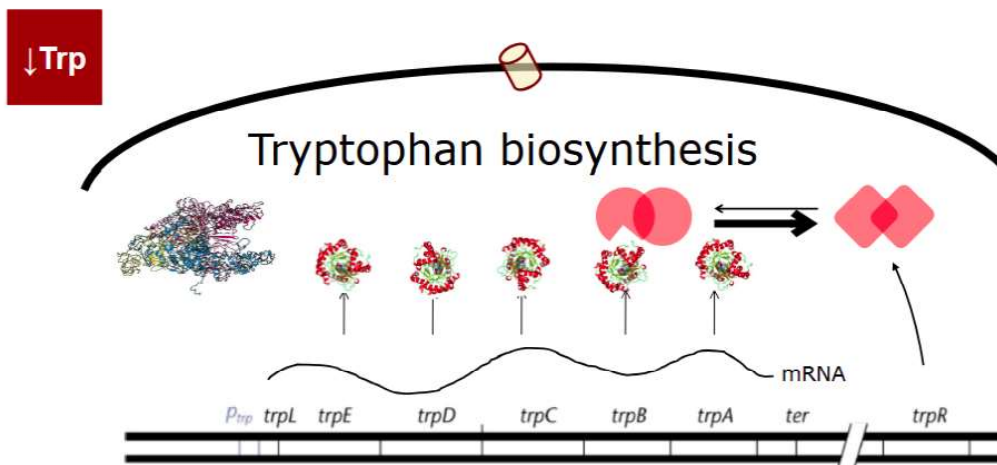
- Transcription Termination/Antitermination

Low trp:

3. Gene regulation by RNA structure

Attenuation – Transcription Termination/Antitermination

Example: The Tryptophane operator in *Escherichia coli*



- The Tryptophane attenuator in *Escherichia coli*

- Riboswitches????

- Attenuators

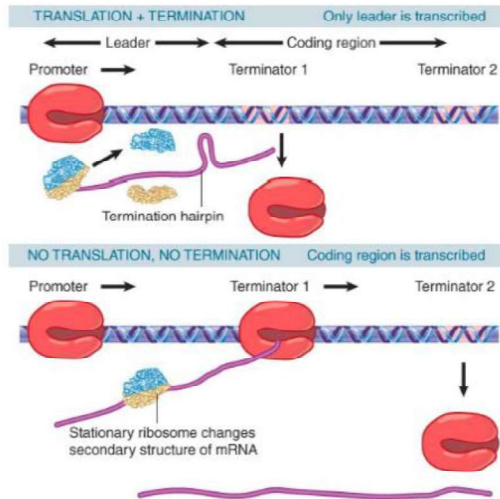
3. Gene regulation by RNA structure

Attenuation – Transcription Termination/Antitermination

Example: The Tryptophane attenuator in *Escherichia coli*



- **Attenuation** – The regulation of bacterial operons by controlling termination of transcription at a site located before the first structural gene.
- An **attenuator** (intrinsic terminator) is located between the promoter and the first gene of the *trp* cluster; ie in the leader.
- The absence of Trp-tRNA suppresses termination and results in a 10^3 increase in transcription.
- The **leader** is the first part of the mRNA that does not encode a protein



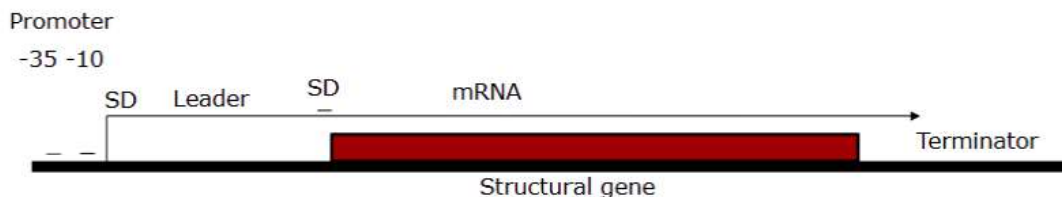
The leader, RNA structure, low or high tryptophan

3. Gene regulation by RNA structure

Attenuation – Transcription Termination/Antitermination

Example: The Tryptophane attenuator in *Escherichia coli*

- The **leader** is the first part of the mRNA at the 5' that does not encode a protein
- The leader sequence of the Trp operon contains two codons for Trp
- The structure of RNA at the attenuator depends on the speed by which this reading frame is translated.
 - $\uparrow$ Trp $\rightarrow$ fast transcription $\rightarrow$ formation of the terminator
 - $\downarrow$ Trp $\rightarrow$ slow transcription $\rightarrow$ formation of the **anti-terminator**



Summary:

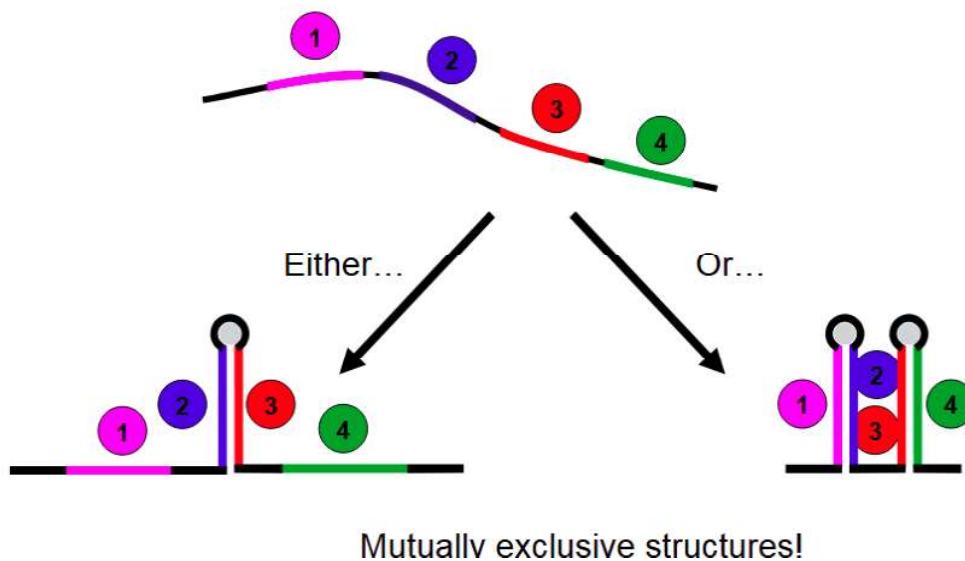
- The leader region of the *trp* operon has a 14-codon open reading frame that includes two codons for tryptophan.
- The structure of RNA at the attenuator depends on the speed by which this reading frame is translated.

- In the presence of Trp-tRNA, the leader is translated fast, and right after the RNA polymerase. The attenuator is able to form the hairpin that causes termination, because the formation of the antiterminator is prevented due to blocking by the ribosome.
- In the absence of Trp-tRNA, the leader is translated slow, there is a gap of naked RNA between the RNA polymerase and the ribosome; hence the anti-terminator is formed causing anti-termination

3. Gene regulation by RNA structure

Attenuation – Transcription Termination/Antitermination

Example: The Tryptophane attenuator in *Escherichia coli*



Either: anti-terminator and or: terminator.

Which structure is preferentially formed? – depends on Trp availability

If we do not have trp: there are two codes for trp but there is none available so the ribosome stops and cannot continue– but the RNA-polymerase will keep going

If we have a lot of tryptophan: the ribosome will detach – so the ribosome is the one controlling whether it will form as the terminator or the antiterminator.

Which genetic element is required for the expression of a gene to be regulated by attenuation?

A transcriptional terminator

###.###%

A leader peptide

###.###%

A Shine-Dalgarno sequence

###.###%

A stop codon

###.###%

- ANSWER: a leader peptide. you always have a transcriptional terminator at the end and there will always be a shine-dalgarno sequence.

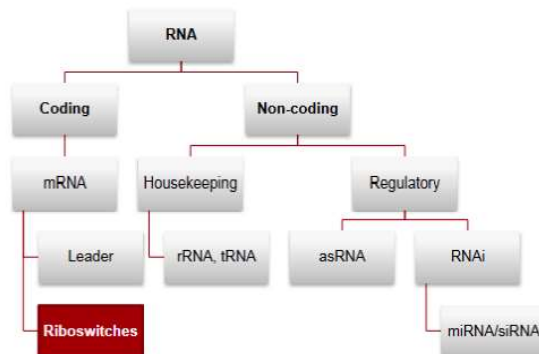
- Riboswitches

Riboswitches are RNA leaders can fold into mutually exclusive structures that either favors or prevents expression of the structural gene

PROKARYOTIC

Riboswitches

- Riboswitches are RNA leaders can fold into mutually exclusive structures that either favors or prevents expression of the structural gene.
- In eukaryotes, control alternative splicing.
- Moreover, they can bind a metabolite.
- The binding of the metabolite stabilizes one of the structures
- Riboswitches may control gene expression at all levels from transcription, over translation to mRNA stability



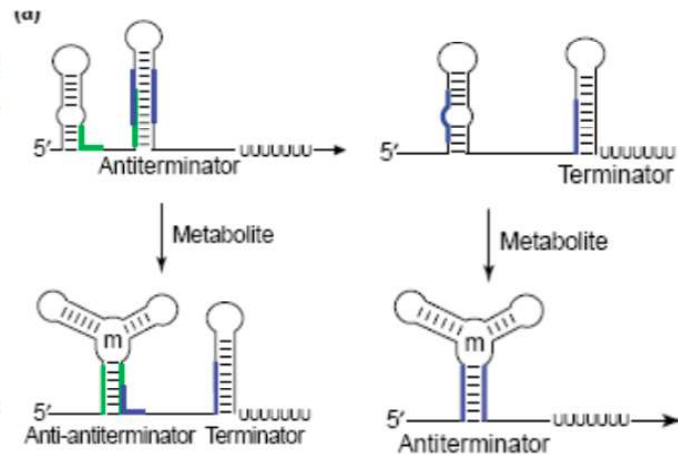
3. Gene regulation by RNA structure

Riboswitches

Attenuators



- Riboswitches are RNA leaders can fold into mutually exclusive structures that either favors or prevents expression of the structural gene.
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- Moreover, they can bind a metabolite.
- The binding of the metabolite stabilizes one of the structures
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<https://web.bii.a-star.edu.sg/~stanley/Riboswitches/The%20riboswitch%20control%20of%20bacterial%20metabolism.pdf>

Translational sequestering

“hiding” the SD sequence – by obscuring or binding

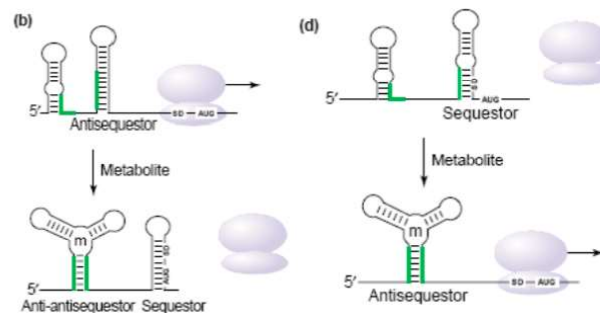
3. Gene regulation by RNA structure

Riboswitches

Translational sequestering



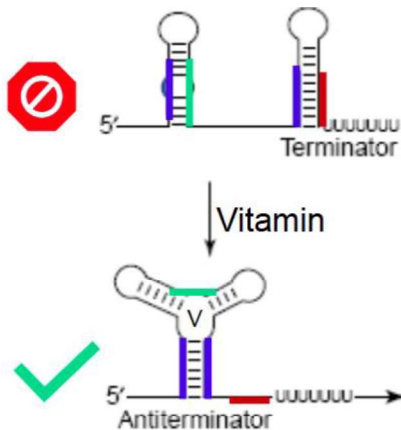
- Sequestering:** avoiding accessibility of the Shine-Dalgarno (SD) sequence, necessary for initiation of translation
- In the presence or absence of a certain metabolite, the SD sequence can be accessible for the ribosome to initiate translation or not upon formation of mutually exclusive structures



<https://web.bii.a-star.edu.sg/~stanley/Riboswitches/The%20riboswitch%20control%20of%20bacterial%20metabolism.pdf>

If a metabolite binds it can change the formation which hides the shine Dalgarno sequence – if the SD sequence is not open (in a single stranded form) it is inaccessible.

Quiz time: A riboswitch based vitamin sensor



You have designed a **vitamin sensor** based on a riboswitch fused to a gene that encodes for a **fluorescent reporter**:

However, the synthesis company made a mistake and **eliminated the red fragment**. You found out when you were making the test experiments in the presence and absence of the vitamin. What did you observe?

- A) In the absence of the vitamin, the fluorescence levels were null
- B) In the presence or absence of the vitamin, you registered high fluorescence levels
- C) In the presence or absence of the vitamin, you registered null fluorescence levels
- D) The vitamin was not able to bind the riboswitch

A riboswitch-based vitamin sensor.

You have designed a vitamin sensor based on a riboswitch fused to a gene that encodes for a fluorescent reporter: However, the synthesis company made a mistake and eliminated the red fragment. You found out when you were doing the test experiments in the presence and absence of the vitamin. What did you observe?

In the absence of the vitamin, the fluorescence levels were null

##.##%

In the presence or absence of the vitamin, you registered high fluorescence levels

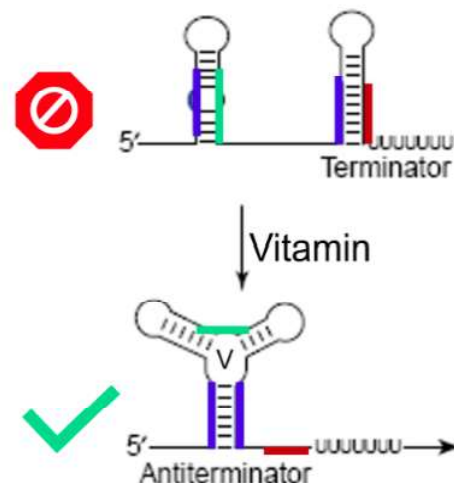
##.##%

In the presence or absence of the vitamin, you registered null fluorescence levels

##.##%

The vitamin was not able to bind the riboswitch

##.##%



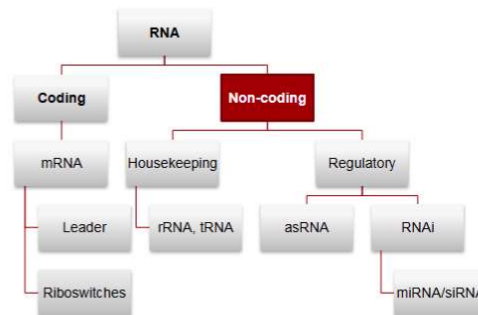
- ANSWER: In the presence or absence of the vitamin, you registered high fluorescence levels. It is a antiterminator so the RNA polymerase will always be able to get to the end and therefore you will observe fluorescence in both cases.

4. Non-Coding RNAs (ncRNA)

4. Non-coding RNAs (ncRNA)



- Bacterial regulator RNAs are called *small RNAs* (**sRNAs**)
- They are also called **antisense RNAs**
- asRNAs can function as regulators by base pairing to specific target RNAs
- **Numerous sRNAs are bound by the protein Hfq, which increases their effectiveness.**

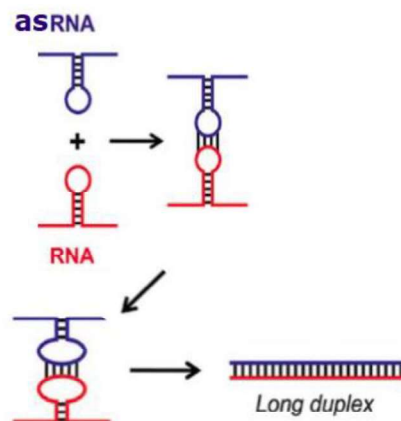


ncRNA^

- Antisense RNA

4. Non-coding RNAs (ncRNA) Antisense RNA (asRNA)

- Bacterial regulator RNAs are called *small RNAs* (**sRNAs**)
- They are also called **antisense RNAs**
- asRNAs can function as regulators by base pairing to specific target RNAs
- **Numerous sRNAs are bound by the protein Hfq, which increases their effectiveness.**



- Very rapidly degraded in any cell because this is how the cells have learned to ? viruses.

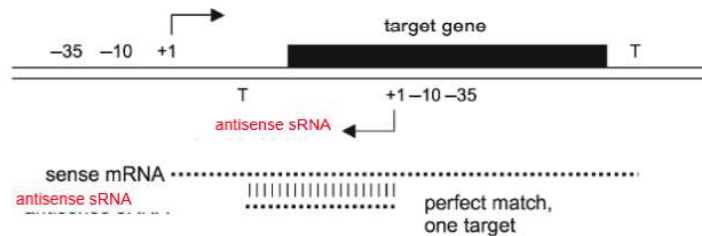
- cis and trans (noncoding) ncRNA antisense RNA (asRNA)

4. Non-coding RNAs (ncRNA)

Antisense RNA (asRNA)

Cis-encoded antisense asRNA

cis-encoded antisense sRNA



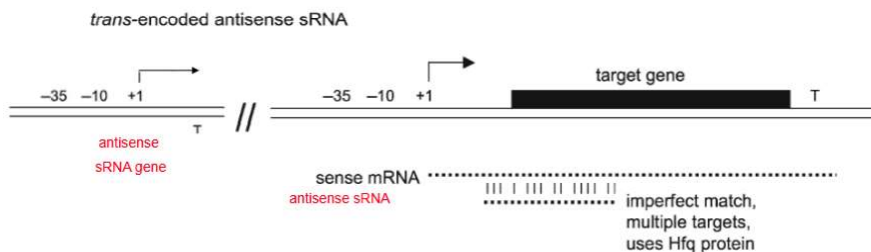
Cis-encoded asRNA are transcribed from the other strand of DNA.
They are a perfect match and are specific for the mRNA target

- cis encoded antisense asRNA: they happen in the same piece of DNA
- trans encoded antisense asRNA: they are incoded in different parts of the genome
 - they are not a perfect match and have multiple mRNA targets.

4. Non-coding RNAs (ncRNA)

Antisense RNA (asRNA)

Trans-encoded antisense asRNA



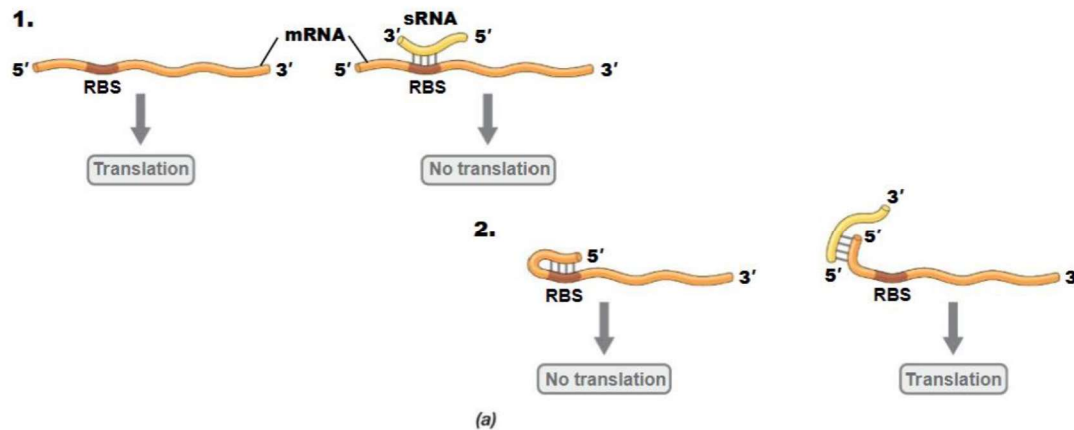
Trans-encoded asRNA are transcribed from a different DNA region
They are a not a perfect match and have multiple mRNA targets

- Shine-Dagarno sequestering

4. Non-coding RNAs (ncRNA)

Antisense RNA (asRNA)

Shine-Dalgarno sequestering/releasing



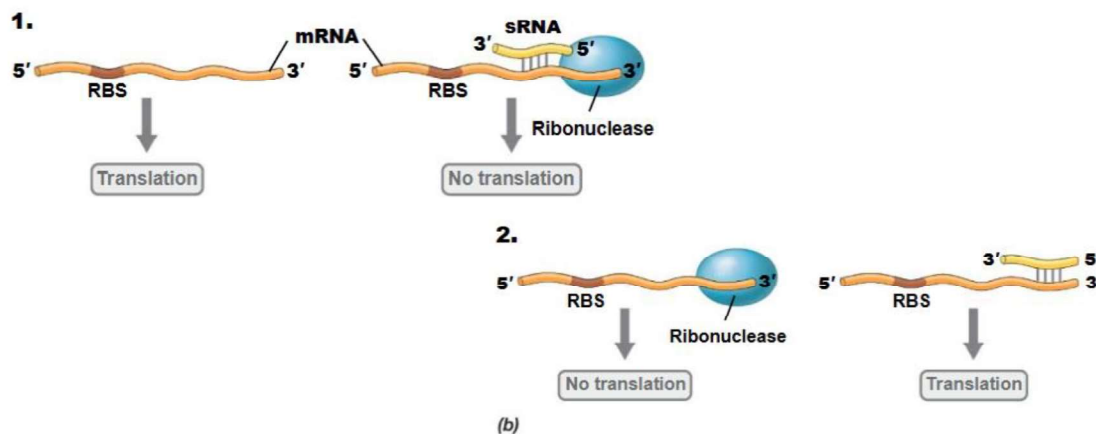
- Regulatory RNAs can both increase and decrease translation

• mRNA de/estabilization

4. Non-coding RNAs (ncRNA)

Antisense RNA (asRNA)

mRNA de/estabilization



- this is the most common one – as soon as there is a piece of double stranded RNA this will occur. The ribonuclease will “start chewing”.
- Will avoid degradation of RNA????

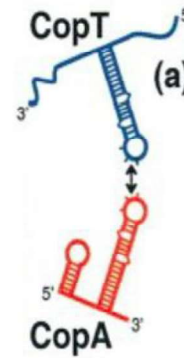
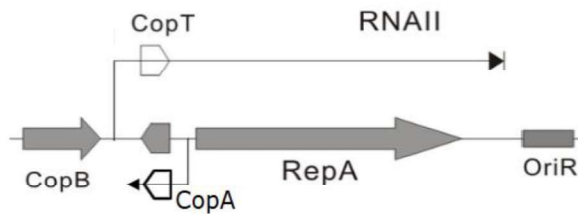
• Unzipping RNA structures: Hfq – kissing complex

- mi/siRNA

4. Non-coding RNAs (ncRNA) Antisense RNA (asRNA)

Unzipping RNA structures: kissing complex

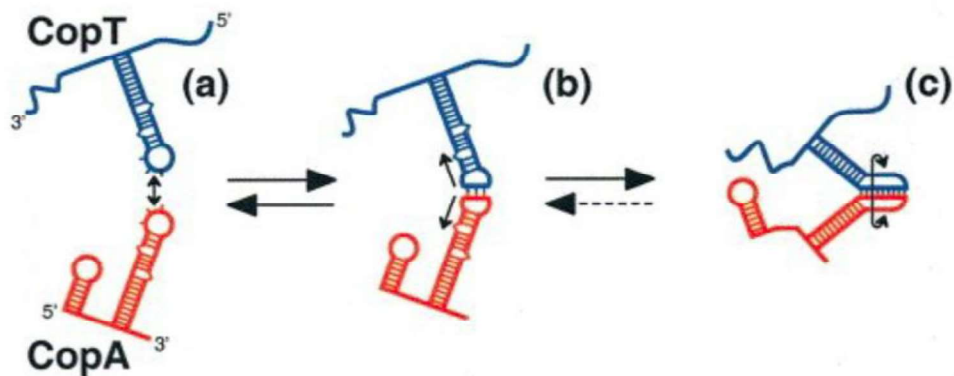
- The concentration of the replication protein (RepA) of the *E. coli* plasmid R1 is controlled by a small RNA (sRNA) CopA
- CopA binds to the RepA encoding mRNA at CopT and prevent it from being translated
- CopA-CopT interaction is initiated in a kissing complex



- cop T and cop A is the same but from each strand (??)

4. Non-coding RNAs (ncRNA) Antisense RNA (asRNA)

Unzipping RNA structures: kissing complex

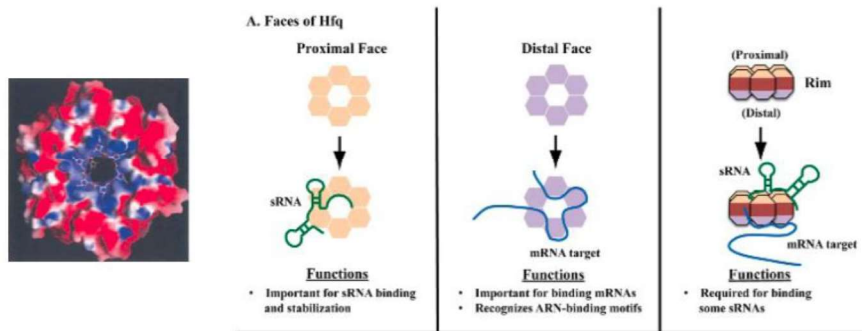


4. Non-coding RNAs (ncRNA)

Antisense RNA (asRNA)

Unzipping RNA structures: Hfq complex

- The Hfq complex is An RNA chaperone, needed for many interactions between trans-sRNA's and their RNA targets

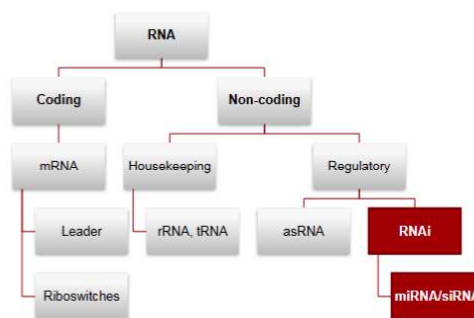


- they allow RNA-RNA interaction in a safe manner – the Hfq
- RNA interference

4. Non-coding RNAs (ncRNA)

mi/siRNA – RNA interference

- Eukaryotic genomes encode many short RNA molecules called *microRNAs* (miRNAs).
- RNA interference (RNAi)** – A process by which short (21 to 23) nucleotide antisense RNAs, derived from longer double-stranded RNAs, can modulate expression of mRNA by translation inhibition or degradation.
- Small interfering RNAs (siRNAs)**, or silencing RNAs
- The nomenclature is a mess...



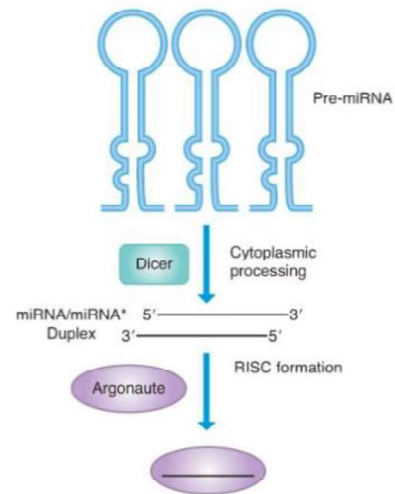
- Eukaryotic genomes
- Silencing RNA (siRNA) means single target, microRNA (miRNA) is multiple targets – the function is the same
- Processing

4. Non-coding RNAs (ncRNA)

mi/siRNA – RNA interference

Processing

- **miRNA** - is first transcribed as a pre-miRNA
- **Drosha** – An endonuclease that processes double-stranded primary RNAs into short, ~70 base pair precursors.
- **Dicer** – An endonuclease that processes double-stranded precursor RNA to 21 to 23 nucleotide miRNA molecules.
- **Argonaute** proteins are binding to the mature siRNA
- **RISC** – RNA-induced silencing complex, a ribonucleoprotein particle composed of a short, single-stranded siRNA and a protein



The Hfq plays a similar role in prokaryotes

- Drosha, miRNA, dicer, argonaute, RISC.
- **Drosha** : An endonuclease that processes double-stranded primary RNAs into short, ~70 base pair precursors – so from pri-miRNA / pre-miRNA to pre-miRNA
- **Dicer** : another nuclease, happens in cytoplasmic processing
- **Argonaute**: has a single strand RNA on it and is ready to bind from the start
- **RISC**: RNA-induced silencing complex

• Function:

- siRNAs and miRNAs

4. Non-coding RNAs (ncRNA)

miRNA – RNA interference

Function

- siRNAs and miRNAs share many similarities, both are short duplex RNA molecules that exert gene silencing effects at the post-transcriptional level by targeting messenger RNA (mRNA), yet their mechanisms of action and clinical applications are distinct. The major difference between siRNAs and miRNAs is that the former are highly specific with only one mRNA target, whereas the latter have multiple targets.
- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4877448/>

• RNAi

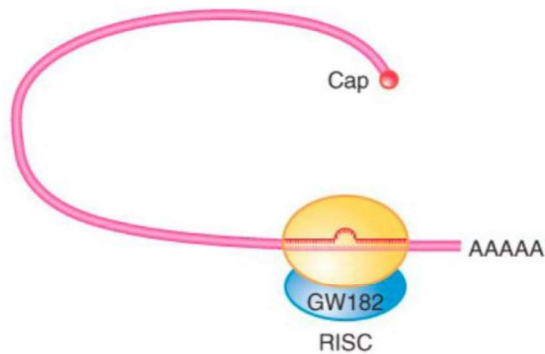
- siRNA → one target
- miRNA → multiple targets

4. Non-coding RNAs (ncRNA)

miRNA – RNA interference

Function

- Argonaute/RISC complexes bind to its target mRNA specified by RNA:RNA base-pairing



- RISC complex – RISC complexes interact with different parts of the initiation
- Inhibition of translation



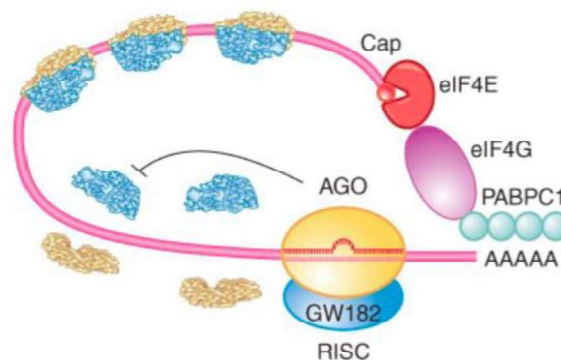
4. Non-coding RNAs (ncRNA)

miRNA – RNA interference

Function: Inhibition of translation

(A) Inhibition of translation elongation

- MicroRNAs (miRNAs; red) repress translation of target mRNAs by blocking translation elongation or by promoting premature dissociation of ribosomes (ribosome drop-off).



- The argonaute is avoiding the assembly of the large subunit of the translation – it avoids that the ribosome is assembled – avoiding translation completely.

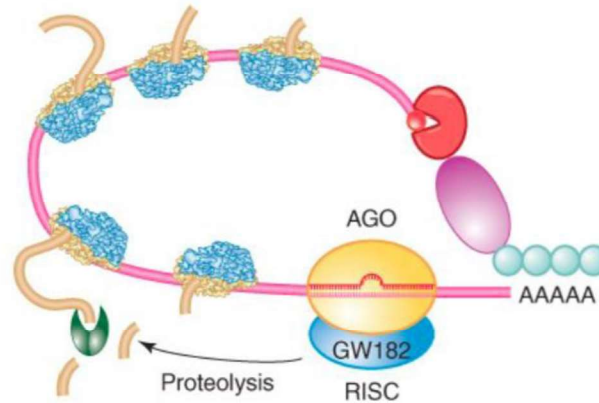
4. Non-coding RNAs (ncRNA)

miRNA – RNA interference

Function: Inhibition of translation

- Co-translational protein degradation.
- This model proposes that translation is not inhibited but rather that the nascent polypeptide chain is degraded co-translationally. The putative protease is unknown

(B) Co translational protein degradation



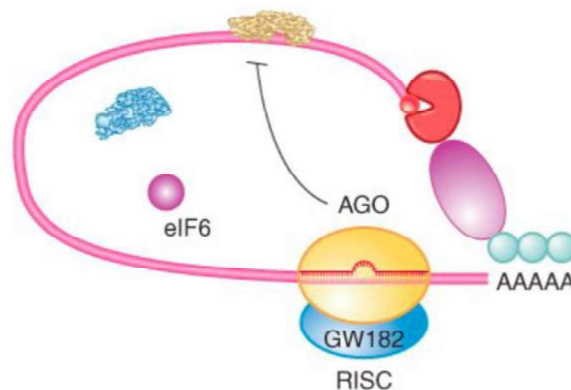
4. Non-coding RNAs (ncRNA)

miRNA – RNA interference

Function: Inhibition of translation

- Argonaute proteins recruit eIF6, which prevents the large ribosomal subunit from joining the small subunit

(D) Inhibition of ribosomal subunit joining



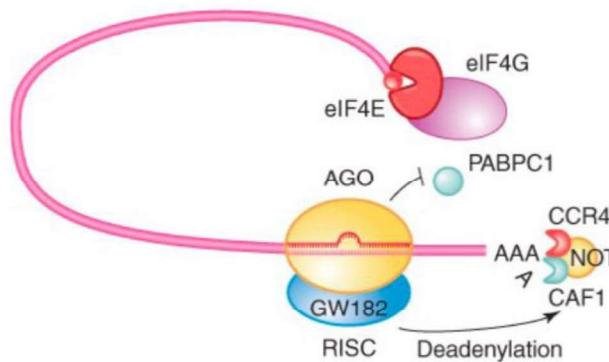
4. Non-coding RNAs (ncRNA)

miRNA – RNA interference

Function: Inhibition of translation

- Argonaute proteins prevent the formation of the closed-loop mRNA configuration by an ill-defined mechanism that includes deadenylation

(E) Inhibition of mRNA circularization through deadenylation



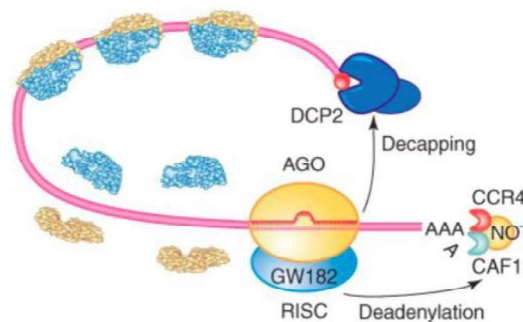
4. Non-coding RNAs (ncRNA)

miRNA – RNA interference

Function: Inhibition of translation

- MicroRNA-mediated mRNA decay. MicroRNAs trigger deadenylation and subsequent de-capping of the mRNA target.

(F) Deadenylation and decapping



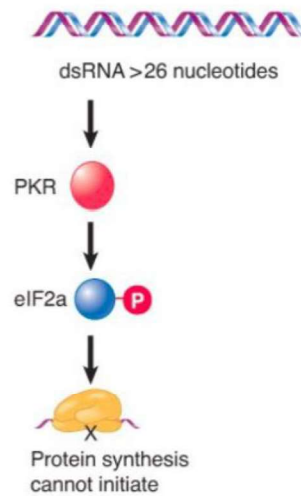
- Defense

4. Non-coding RNAs (ncRNA)

miRNA – RNA interference

Function: defense

- dsRNA may cause silencing of host genes (RNA silencing)

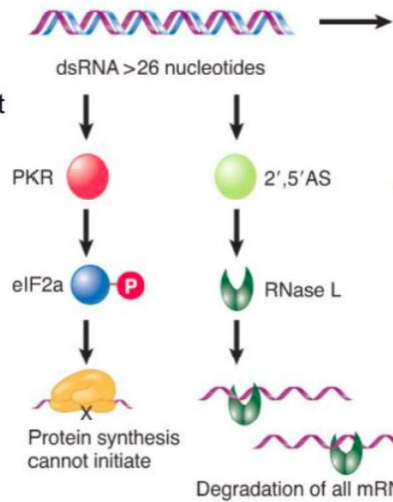


4. Non-coding RNAs (ncRNA)

miRNA – RNA interference

Function: defense

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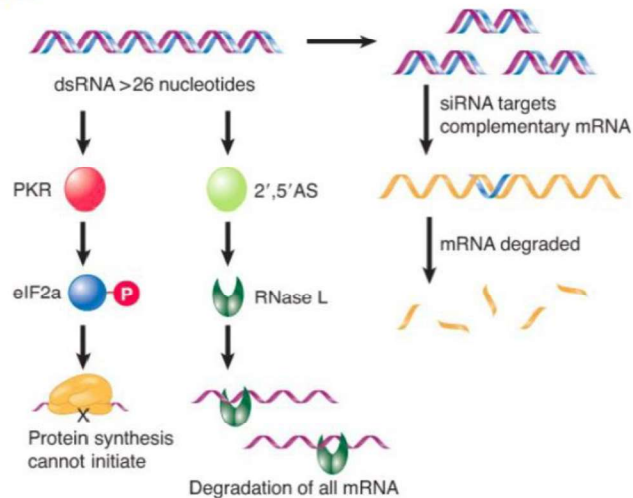


4. Non-coding RNAs (ncRNA)

miRNA – RNA interference

Function: defense

- dsRNA may cause silencing of host genes (RNA silencing)
- General reduction of translation



- The size of the RNA is going to decide whether it is going to shut translation down completely or start degrading the mRNA. – so big dsRNA will result in silencing of RNA – general reduction of the translation.

What elements an RNA should have to go through translation?:

Which statement about bacterial trans-encoded antisense sRNA is TRUE

They have a perfect match to their target

###.###%

They are transcribed from the same promoter as their target

###.###%

They may regulate several genes

###.###%

They are independent of the Hfq protein complex

###.###%

- ANSWER: They may regulate several genes. They are encoded somewhere else, so it is not perfectly matched to their target. They can bind imperfectly to multiple ones.

Which of the following elements is a nucleo-protein complex

Dicer

###.###%

RISC

###.###%

Drosha

###.###%

Argonaute

###.###%

- ANSWER: RISC

Molecular Biology Week 8:

Learning objectives

Notes from the lecture

Molecular Biology Week 9:

Learning objectives

Chapters 10 & 11 (and selected sections in Chapters 7 & 14):

After the lectures and exercises you will understand:

- Replication basics
- The structure of DNA Polymerase
- Replication process basics
- That DNA polymerase has one polarity
- That DNA polymerase needs a primer
- How problems caused by biochemical properties of DNA polymerase are solved
- The structure of a replication fork
- Replication fork dynamics – basics
- The concept of leading and lagging strands during DNA replication
- That Okazaki fragments are formed on the lagging strand
- DNA repair
- How does replication initiate - When to replicate
- One round of replication (and only one) is required to replicate a chromosome
- Replication fork in detail
- Special problems concerning replication of linear chromosomes
- Fidelity of DNA replication

Notes from the lecture

- Lecture 1: DNA Polymerases & Replication Problems Part 1: Need for RNA Primers
- Lecture 2: DNA Polymerases & Replication Problems Part 2: DNA Synthesis Polarity at the Replication Fork
- Lecture 3: DNA Polymerases & Replication Problems Part 3: DNA Synthesis Polarity at Chromosome Ends
- Lecture 4: Replicating Genomes

- Lecture 5: DNA Maintenance - Replication induced Mutations
- Lecture 6: Replication based Biotech Tools

Molecular Biology Week 10: DNA repair and recombination / Recombination and DNA DSB Repair

Learning objectives

Understand what types of DNA Damage is encountered by the Cell

- Understand Basic Repair Principles
- Understand that DNA Double Strand Breaks (DSBs) are the most serious DNA lesions
- Understand that Linear Chromosomes need to be protected from DNA DSB repair
- Obtain Basic Insights into DNA DSB Repair by Non-Homologous End-Joining
- Obtain Basic Insights into DSB Repair by Homologous Recombination
- Understand Recombination based DNA Double Strand Break Repair
- Understand Gene Conversion
- Understand induced Meiotic Homologous Recombination
- Understand Mitotic DNA DSB Repair (Synthesis Dependent Strand Annealing)
- Understand the Functions of key Recombination Protein

Notes from the lecture

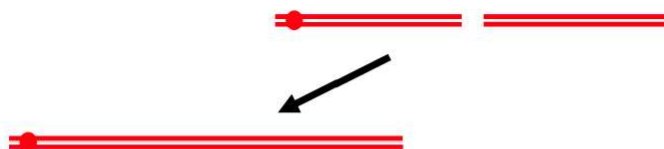
- Lecture 1: Introduction to DNA Double Strand Breaks (DSB) Repair

A mismatch of nucleotides does not form at normal basepair (fx C-A, G-T)

DNA DSB stands for **DNA Double-Strand Break**.

A double-strand break is a type of DNA damage where **both strands** of the DNA helix are severed. This is a serious form of damage that cells must repair to maintain genome stability.

Dirty ants – damaged nucleotides fx if the break is due to radiation der will be lesions that need to be removed (in context of the MRX complex for end joining after DNA double-strand break by non-homologous end-joining)



Non-Homologous End-Joining

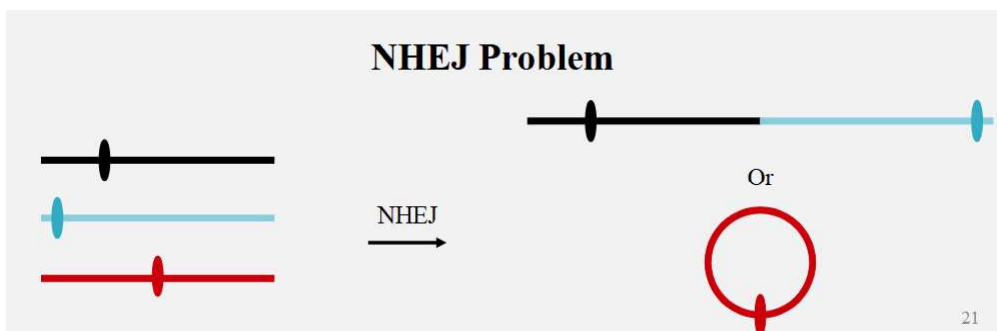
NHEJ problem – RISK of translocations



This makes a big risk as it happens in a chromosome and if it is a sex-chromosome it can affect the sexual ___ of a child. So we need to trap the “ants” together so that this will not happen (the ants being the two smaller parts of the DNA after the double-strand break.)

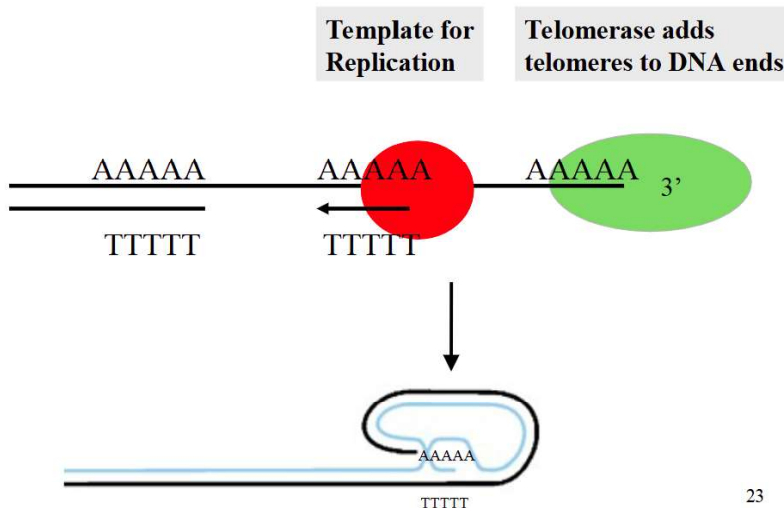
Eukaryotes has linear chromosomes and the ends of linear chromosomes will constantly shrink during generations of replication, which means that at some point it will no longer have any telomers.

Also for the non-homologous end-joining the chromosomes has no telomers (this is fixed by the addition of telomers by telomerase which synthesizes large amounts of ssDNA telomeric repeats like AAAAA and TTTT):



This is why our cells reach a point where they are not so viable anymore.

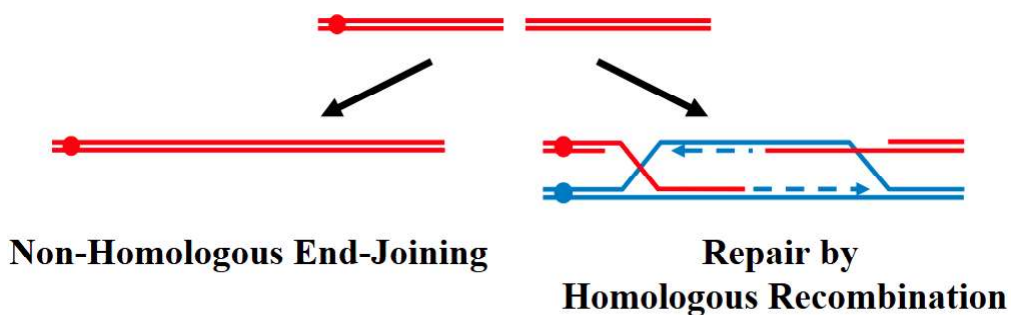
Chromosome-Ends mimick DNA DSBs - Problem solved by forming T-Loop



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To avoid fusing the chromosomes to protect the ants so they are not the subject for DNA repair. The principle: hide the ants so they are not accessible for the repair and we are all good.

Repair by homologous recombination: If a chromosome has been replicated it is composed by two similar strands (sister strands) so to repair a DNA DSB we can use the sister strand, as long as it is a diploid organism, you can pair the homologous chromosomes as a template. (but this is a bad idea??)



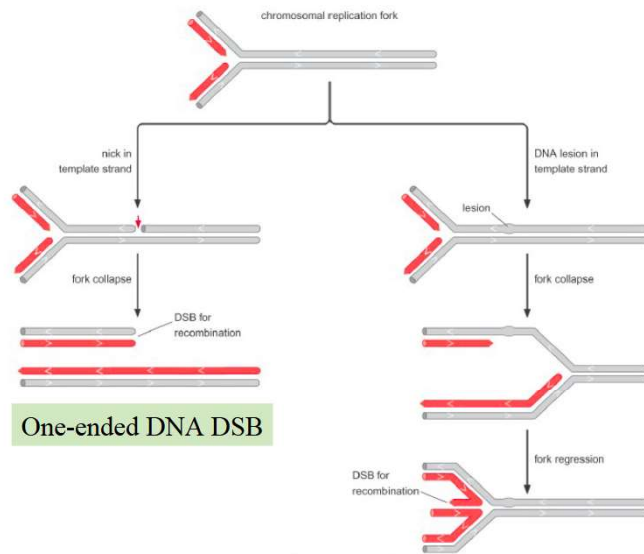
So Non-homologous end-joining is very yolo, we just join them and that is it, no matter if it is the wrong ants that is paired.

Repair by homologous recombination we check the ants if they belong together before joining them.

If both ants are able to base-pair on both sides then there is a big chance that they can bind.

So you use the sister sequence to look if the ends are similar to the sister sequence and thereby see if the two ants belong together or not. (if they do not then they do not belong together).

Collapsed Replication Forks are rescued by Homologous Recombination A One-ended DNA Double Strand Break.



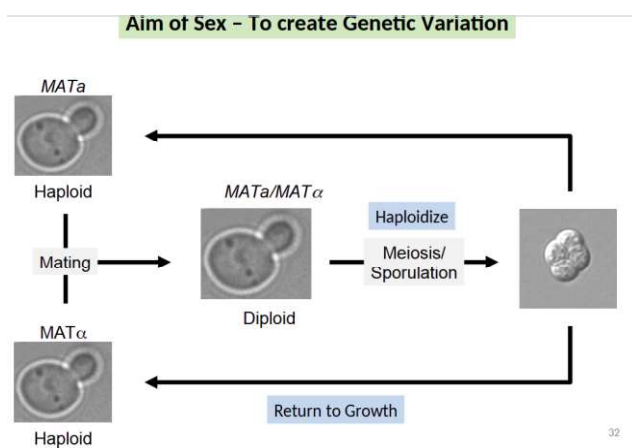
This happens if there exists a DNA-nick that is not repaired before replication – we indirectly get a DNA DSB. (DNA double-stranded break)

You can recruit DNA helicases stripping off the ends from the right to the left – so one strand get the full replication and then the other strand finish after the lesion and uses this strand as a template instead of the original one with the lesion. (fork regression)

• Lecture 2: Meiotic Homologous Recombination – Part 1

“Put the ends together and then test them before joining them”

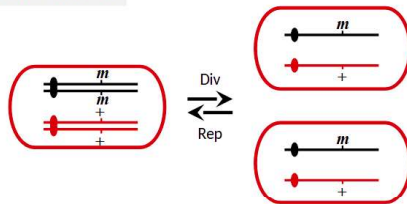
Sexual cycle of bakers yeast:



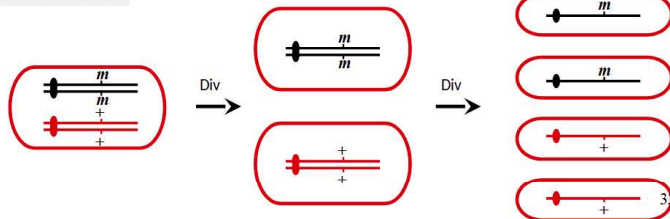
Mitotic and Meiotic Divisions:

Hallmark here is that the cells are the same – identical – mitotic division.

Mitotic Division

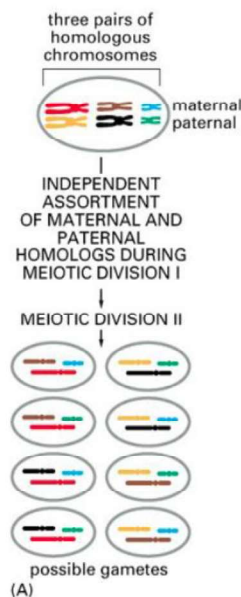


Meiotic Division



For meiotic division we reduce the number of chromosomes in the cell – haploid gametes combine to form diploids – do you either get a chromosome from your father or your mother.

Meiosis Contributes in Two Ways to Create Genetically Different Gametes:



In this example:

- 8 different haploid gametes
- 64 different diploids

In Bakers yeast – 16 Chr:

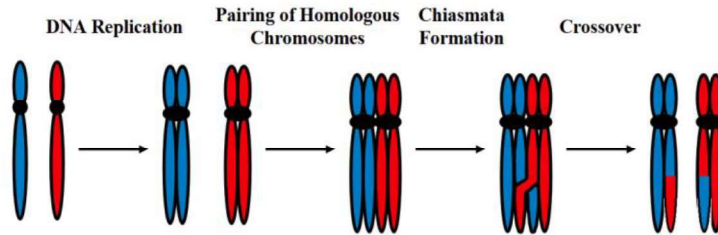
- 65.536 different haploid gametes
- 4.294.967.296 different diploids

2ⁿ different haploid gametes

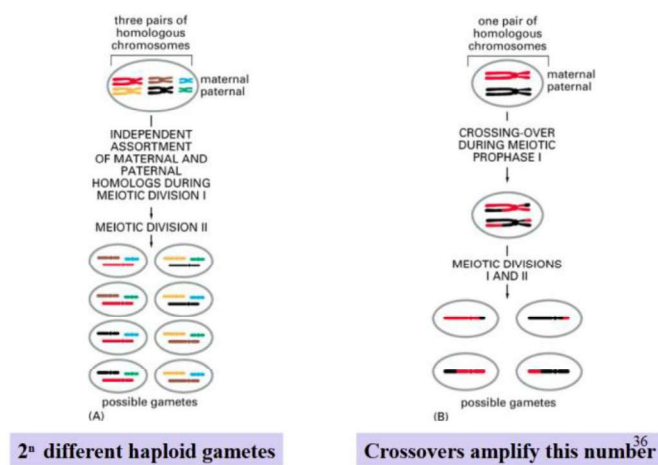
You can get 64 different diploids out of this system with 8 different haploid gametes. It quickly goes up as it grows at the power of n.

Homologous Chromosomes are Crossing Over in Meiotic Division:

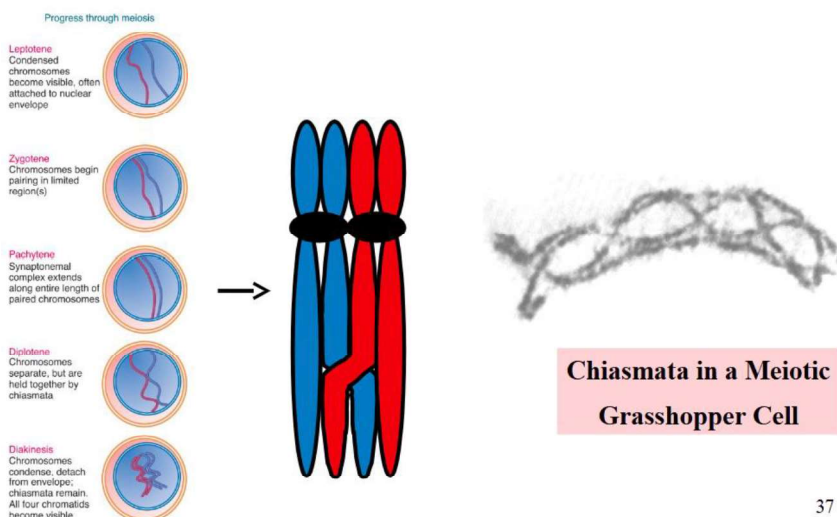
Homologous Recombination



The crossing over of the chromosomes will result in even more room for evolution and adaption in a changing environment.



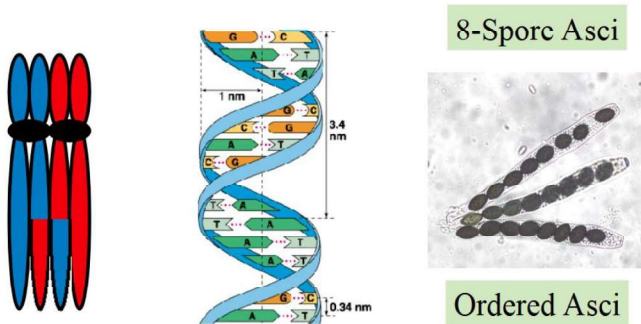
Crossing Over of chromosomes happens in the 1. Meiotic Division:



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Meiosis in Baker's Yeast results in Asci with Four Spore – this is used to show that gene Conversion is a Non-Conservative Event in Meiosis. (search slides)

Meiosis in some Filamentous Fungi results in Asci with Eight ordered Spores:



8-Spore Ascus

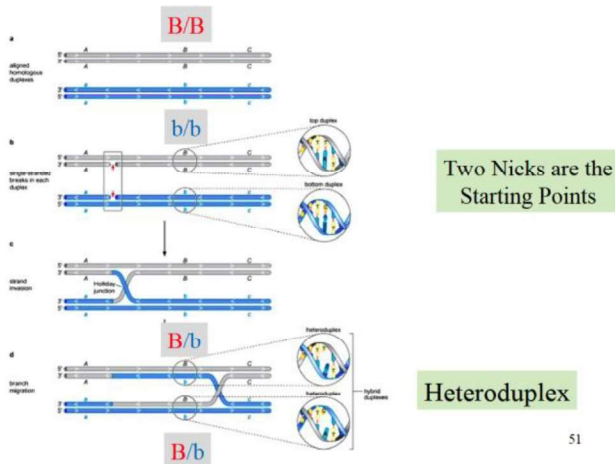
Ordered Ascus

- Crossing over is a conservative event: 4:4 (A vs a gene)
- Gene conversion is a non-conservative event: 5:3 and 6:2 are occasionally observed (A vs a gene)

Lecture 3: Meiotic Homologous Recombination – Part 2

“How does it happen with the strands” – I think he means how the crossover happens.

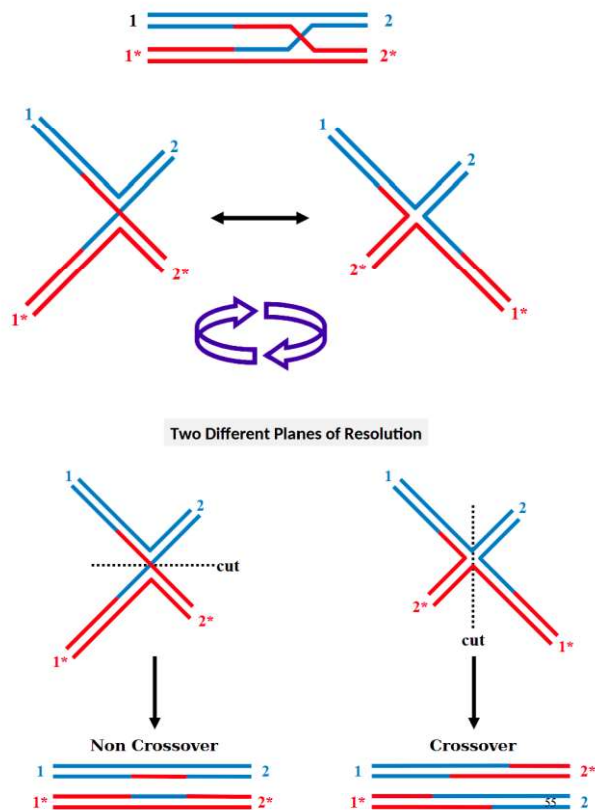
The Holliday Junction – Branch Migration:



Basically explains how you can transfer information from one molecule to another one – so that x can slide back and forth creating the x form and breaking it again.

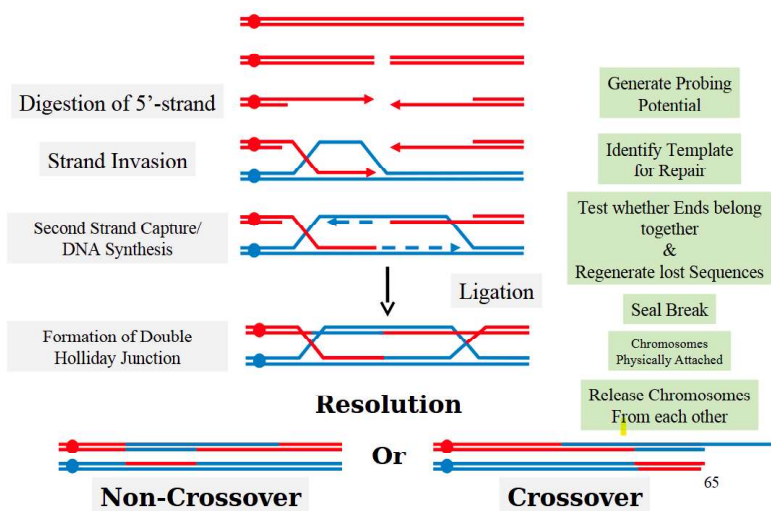
Two nicks – a wrong nucleotide – the cell decides that there is a lesion and it can repair it so the convert one allele to another. – OBS No one found these nicks and there is no enzyme for making these nicks – but people came up with that it cannot be nicks, but it could be breaks.

Isomerization of a Holliday Junction:

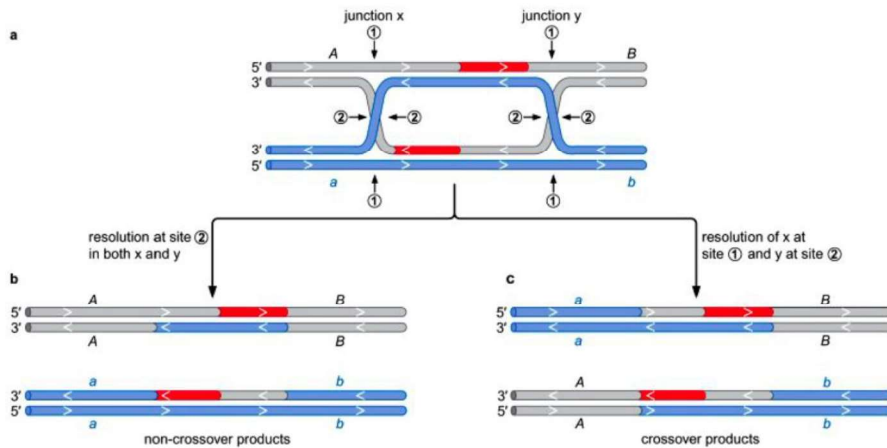


50% of the events should by this theory have a non crossing over and the 50% should be crossover and that in fact is the case so yay for Holliday.

A DSB Repair Model for Homologous Recombination – Meiosis: so if it is breaks and not nicks that start it off:

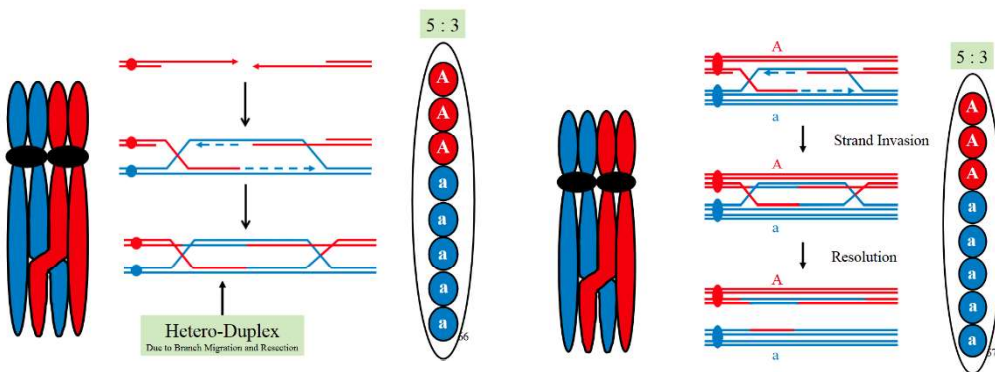


Resolution of a Double Holliday Junction:

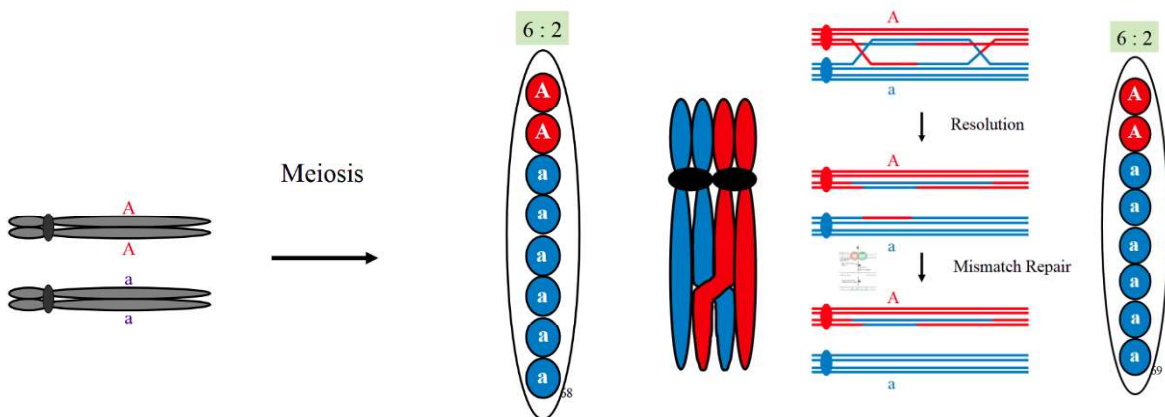


Junction 1 and 1 is a double negative and will result in no crossing over, but junction 1 and then 2 will result in a crossing over this given the data that it is 50/50 yet again.

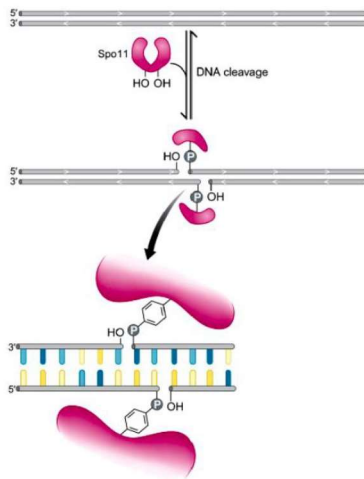
There is two corresponding strands that did not participate in this, we need to be aware of this:



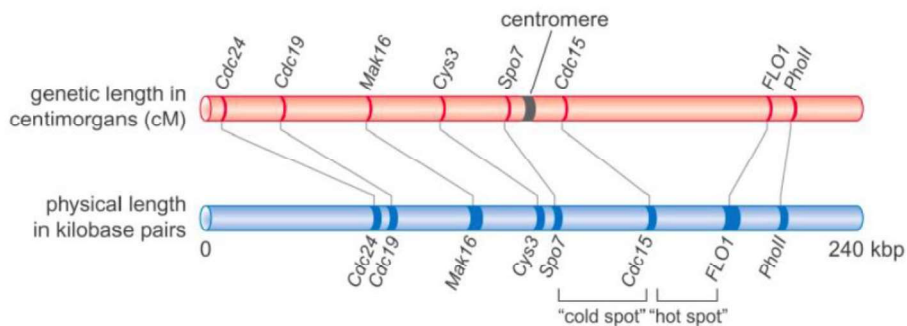
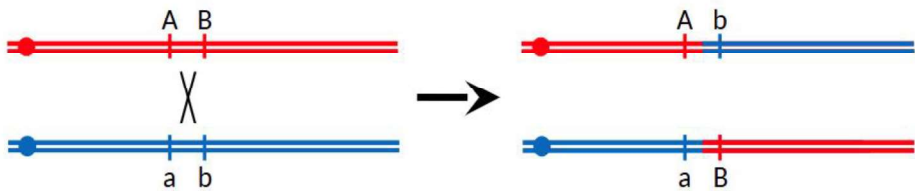
Remember this when we're talking about the spores (because this whole thing was about phenotypes of bakers yeast, whether we see 4:4 of blue and red or we see 5:3 or 6:2).



Meiotic DSBs are induced by Spo11 - a Topoisomerase Type II Homolog (how the breaks are made):

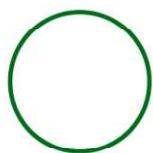


Chance of a Crossover between two Alleles increases with Distance – Centimorgans:



Prokaryotes vs eukaryotes chromosomes:

Prokaryotes



- Does not support a Sexual Division Cycle
- No Biological Timer

Eukaryotes



Problems:

- Completion of Replication – Session 3
- DNA Damage Mimick – Session 4
- And what about Topology? – Session 2